

Discrete Response, Time Series and Panel Data - Exercises

Nils Rechberger

2026-03-05

Series 1

Exercise 1.1

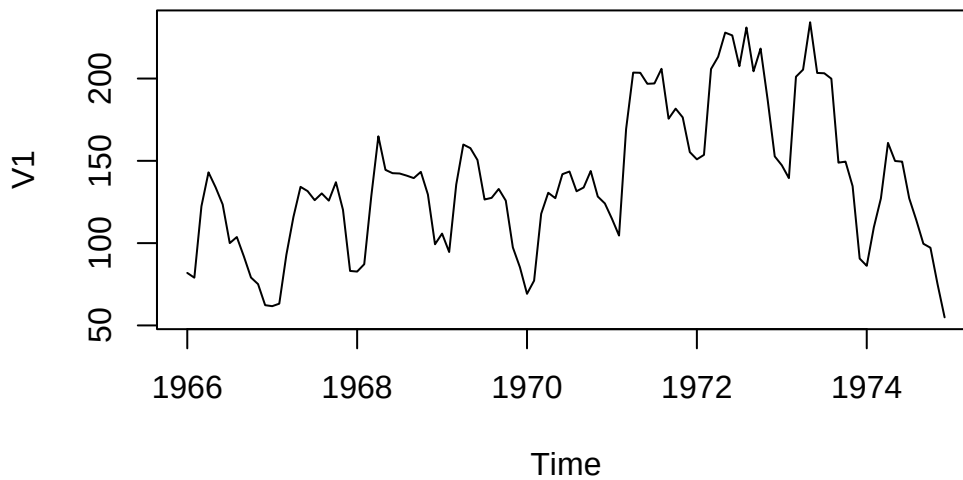
What is the expected period (time period of repetition) and the time step for the following timeseries:

- Sunshine duration per month in Basel from 1990 to 2000: Timestep = 1 month, Period = 12 months.
- Number of newborn babies in the city of Zurich per year from 2000 to 2011: Timestep = 1 year, Periode = None.
- Number of reservations in a restaurant for every night during 4 weeks: Timestep = 1 day, Periode = 7 days.
- Water runoff of a river. The data has been collected every day for 4 years: Time step = 1, periode = 365 days.

Exercise 1.2

```
hstart <- read.table("./data/hstart.dat")  
  
ts <- ts(hstart, start=c(1966,1), freq=12)  
  
plot(ts, main="Residential construction in the USA")
```

Residential construction in the USA



This is not a stationary time series. A stationary time series is one whose statistical properties (mean, variance, etc) are constant over time. This data fails that definition due to two distinct types of non-stationarity.

- Trend: The long-term “level” of the data shifts upward and then downward.
- Seasonality: There are recurring annual cycles.
- Changing Variance: The intensity of the fluctuations changes over the years.

Exercises 1.3

A

```
set.seed(1)

Et <- ts(rnorm(101, 0, 1))

Et[1] <- 0
y1 <- 0 #delete later

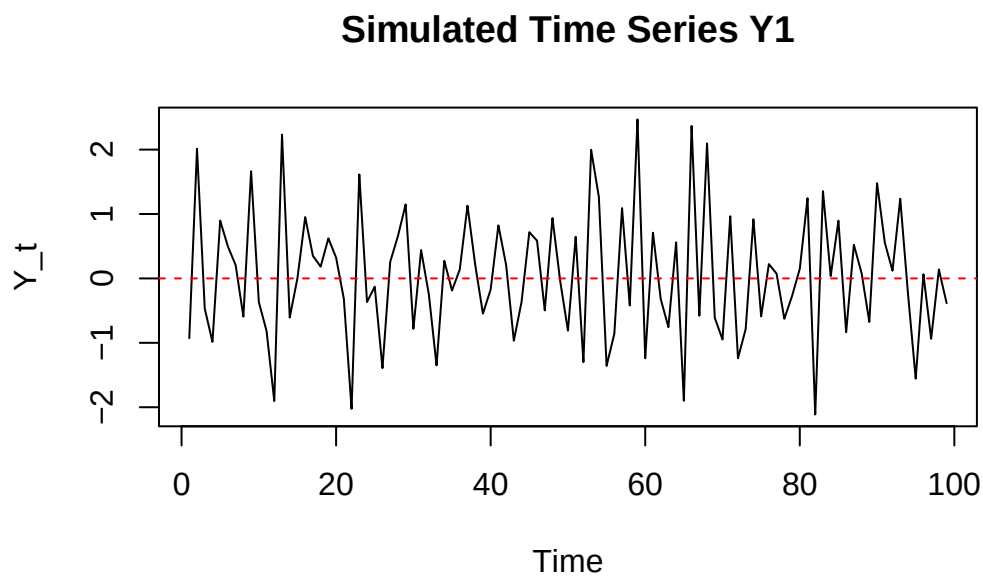
for (i in 2:length(Et)) {
  y1[i-1] <- Et[i] - 0.5 * Et[i-1]
}
```

```

y1 = y1[2:length(y1)]
ts.y1 = ts(y1)

plot(ts.y1, main="Simulated Time Series Y1", ylab="Y_t", xlab="Time")
abline(h=0, col="red", lty=2)

```



The time series seems to be stationary.

B

```

set.seed(1)

Et <- ts(rnorm(101, 0, 1))

Et[1] <- 0
y1 <- 0 #delete later

for (i in 2:length(Et)) {
  y1[i] <- y1[i-1] + Et[i]
}

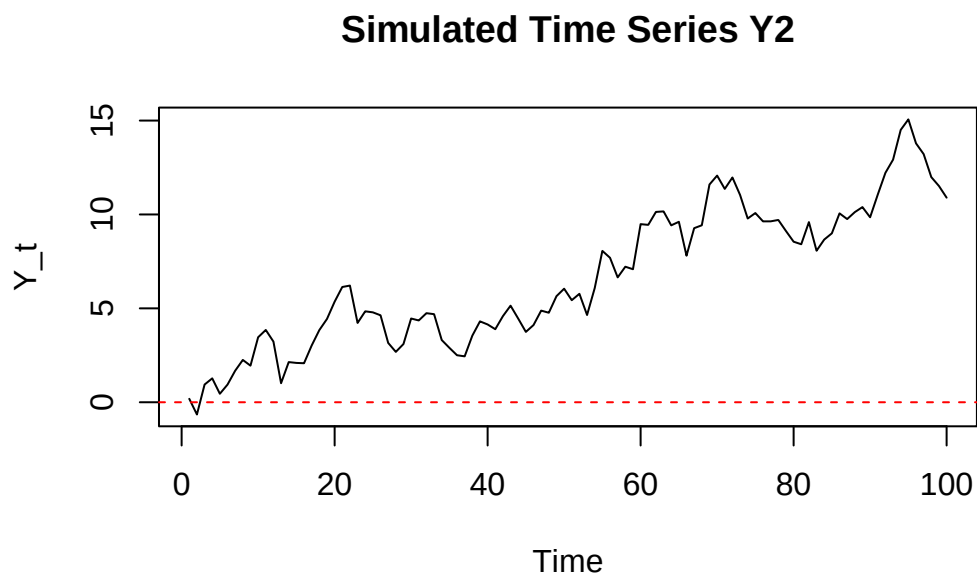
```

```

y1 = y1[2:length(y1)]
ts.y1 = ts(y1)

plot(ts.y1, main="Simulated Time Series Y2", ylab="Y_t", xlab="Time")
abline(h=0, col="red", lty=2)

```



The time series seems not to be stationary.

C

```

set.seed(1)

Et <- ts(rnorm(101, 0, 1))

Et[1] <- 0
y1 <- 0 #delete later

for (i in 2:length(Et)) {
  y1[i] <- 0.5 * y1[i-1] + Et[i]
}

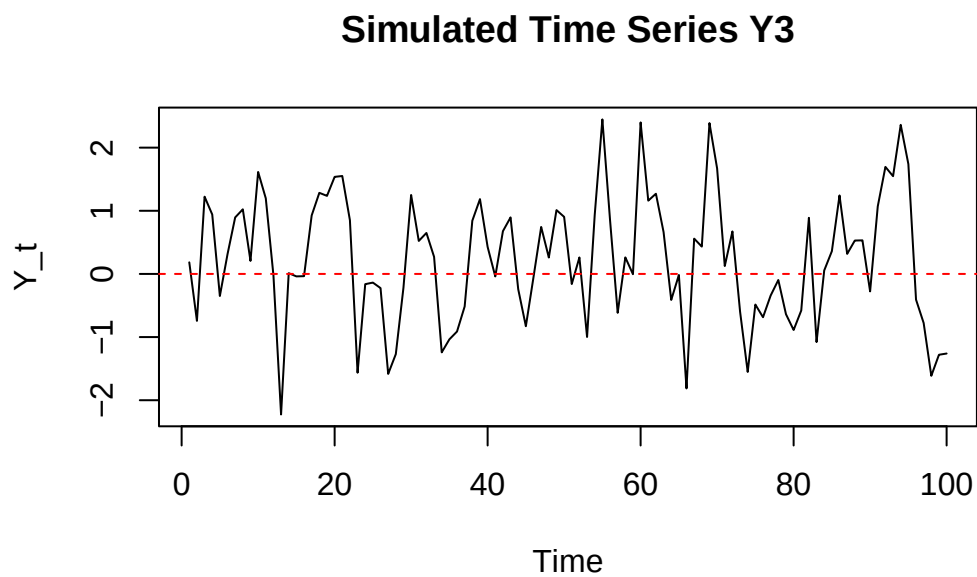
```

```

y1 = y1[2:length(y1)]
ts.y1 = ts(y1)

plot(ts.y1, main="Simulated Time Series Y3", ylab="Y_t", xlab="Time")
abline(h=0, col="red", lty=2)

```



The time series seems to be stationary.

D

```

set.seed(1)

Et <- ts(runif(101, 0.95, 1.05))

Et[1] <- 0
y1 <- 1 #delete later

for (i in 2:length(Et)) {
  y1[i] <- y1[i-1] * Et[i]
}

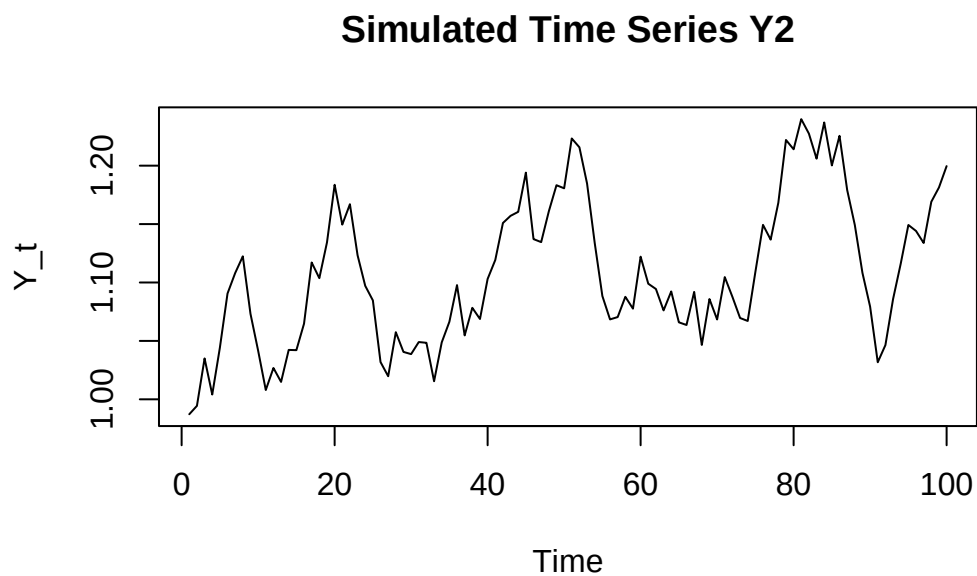
```

```

y1 = y1[2:length(y1)]
ts.y1 = ts(y1)

plot(ts.y1, main="Simulated Time Series Y2", ylab="Y_t", xlab="Time")
abline(h=0, col="red", lty=2)

```



The time series seems not to be stationary.

Series 2

Exercise 2.1

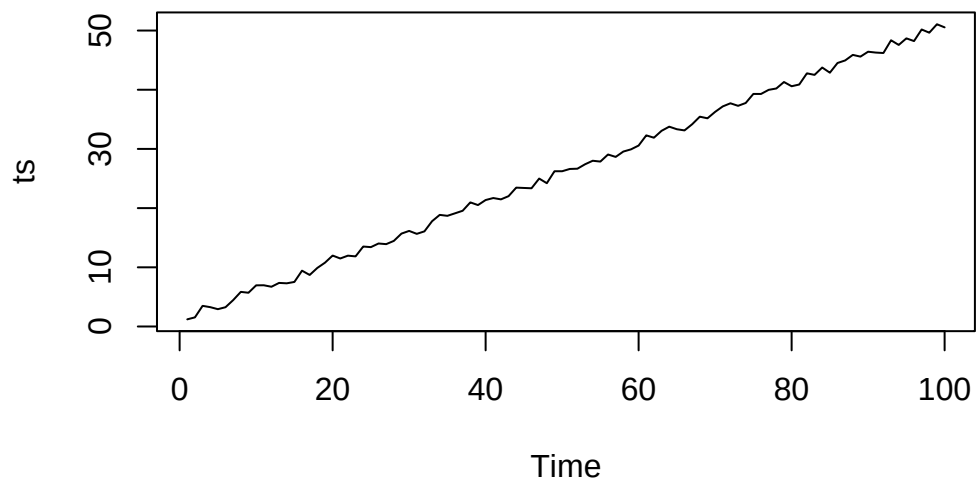
A

```

# create timeseries object
t <- seq(1, 100, length = 100)
data <- 0.5 * t + 1 + runif(100, -1, 1)

ts <- ts(data)
plot(ts)

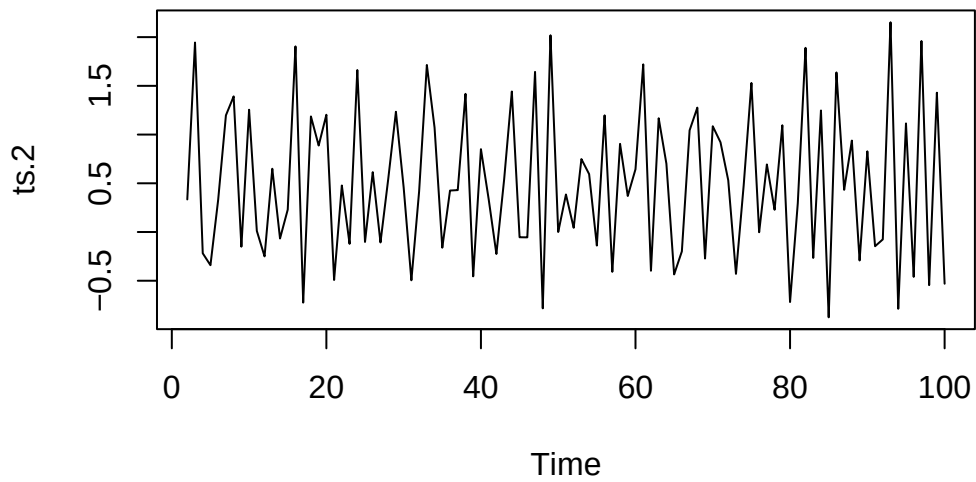
```



```
length(ts)
```

```
[1] 100
```

```
# Applying backward differencing  
ts.2 <- diff(ts, lag = 1)  
plot(ts.2)
```



```
length(ts.2)
```

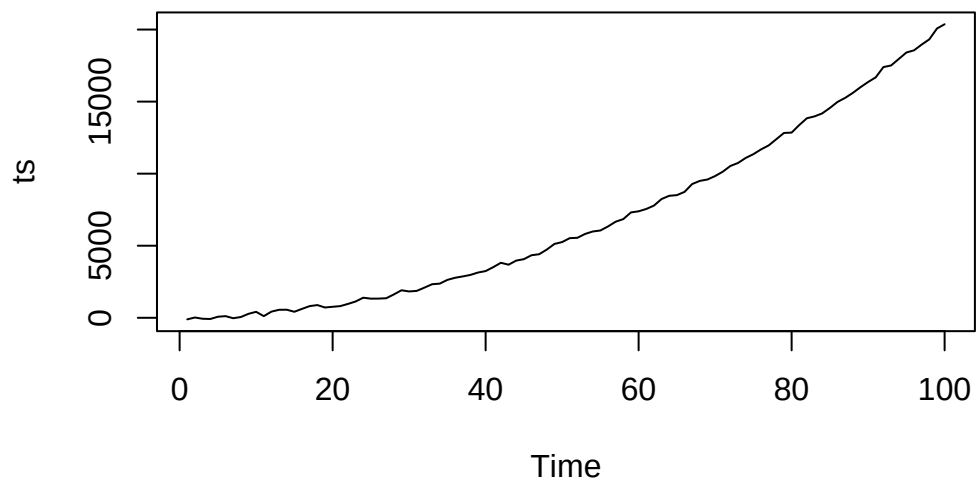
```
[1] 99
```

The new time series are 99 units long instead of 100.

B

```
# create timeseries object
t <- seq(1, 100, length = 100)
data <- 2 * t **2 + 3 * t - 1 + runif(100, -200, 200)

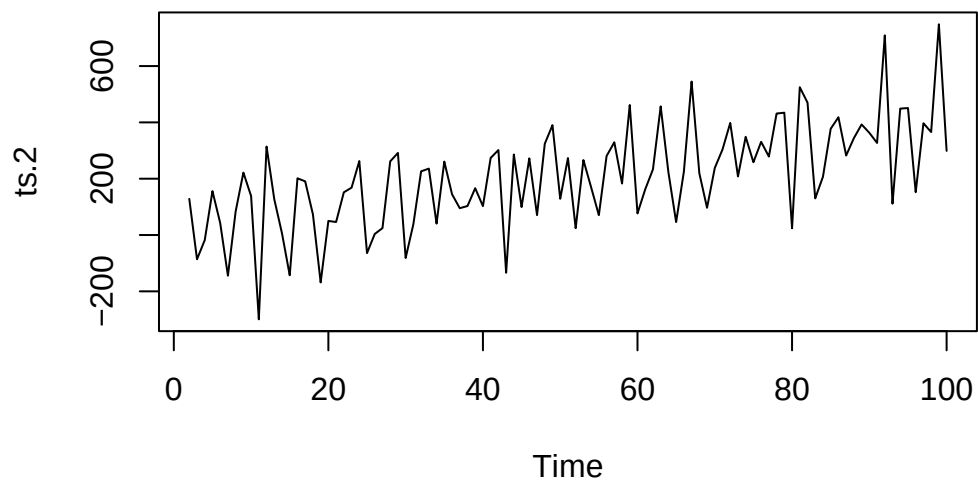
ts <- ts(data)
plot(ts)
```



```
length(ts)
```

```
[1] 100
```

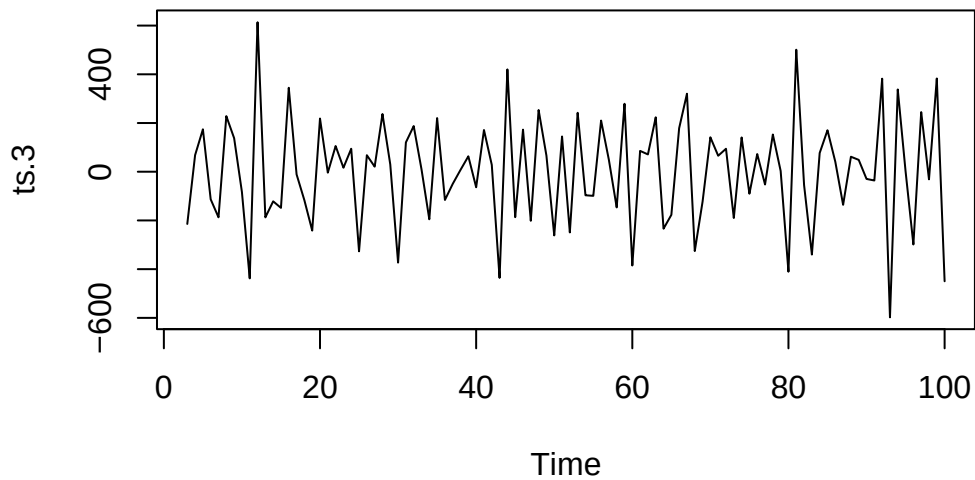
```
# Applying backward differencing  
ts.2 <- diff(ts, lag = 1)  
plot(ts.2)
```



```
length(ts.2)
```

```
[1] 99
```

```
# Again Applying backward differencing  
ts.3 <- diff(ts.2, lag = 1)  
plot(ts.3)
```



```
length(ts.3)
```

```
[1] 98
```

Since we applied backward differencing twice, the length of the output time series increased to 98.

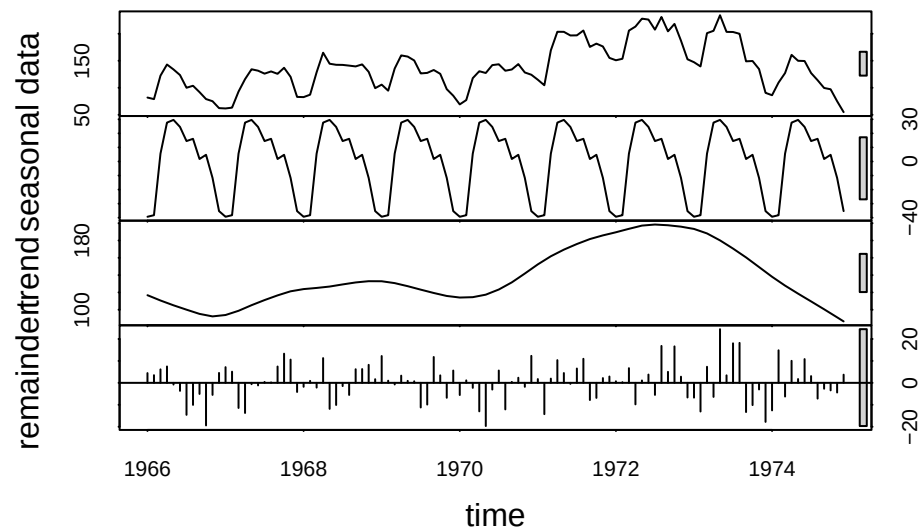
Exercise 2.2

A

```
hstart <- read.table("./data/hstart.dat")

ts <- ts(hstart[, 1], start = c(1966, 1), frequency = 12)
hstart_stl <- stl(ts, s.window = "periodic")

plot(hstart_stl)
```



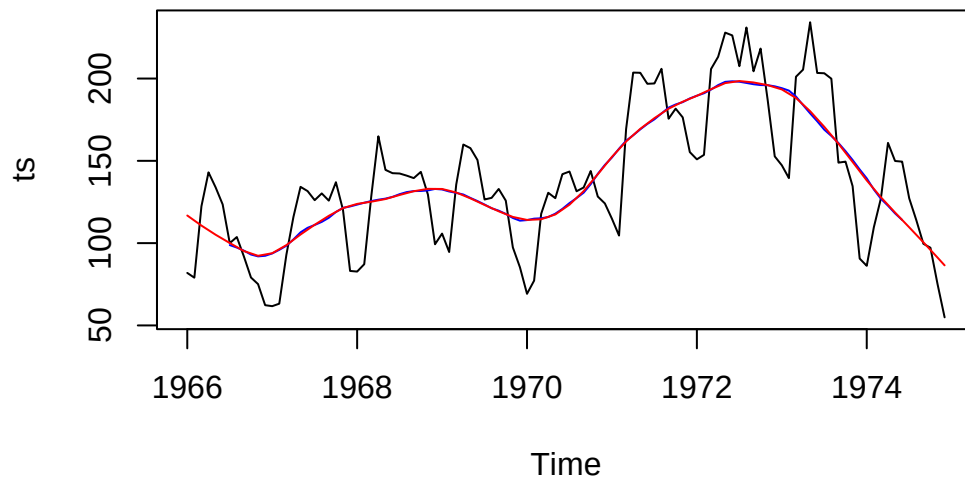
B

```
hstart <- read.table("./data/hstart.dat")

ts <- ts(hstart[, 1], start = c(1966, 1), frequency = 12)

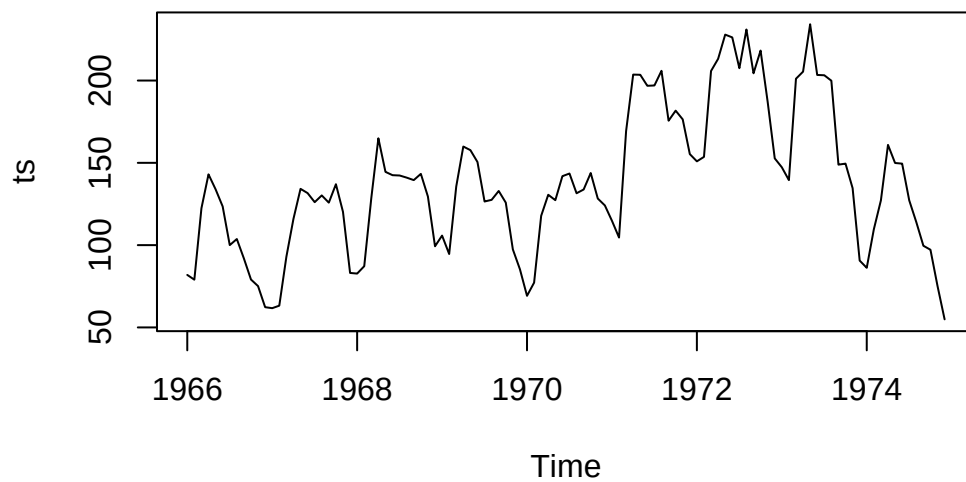
weights <- c(1, rep(2, 11), 1) / 24
filtered <- filter(ts, filter = weights)

plot(ts)
points(filtered, type = "l", col = "blue")
lines(hstart_stl$time.series[, "trend"], col = "red")
```

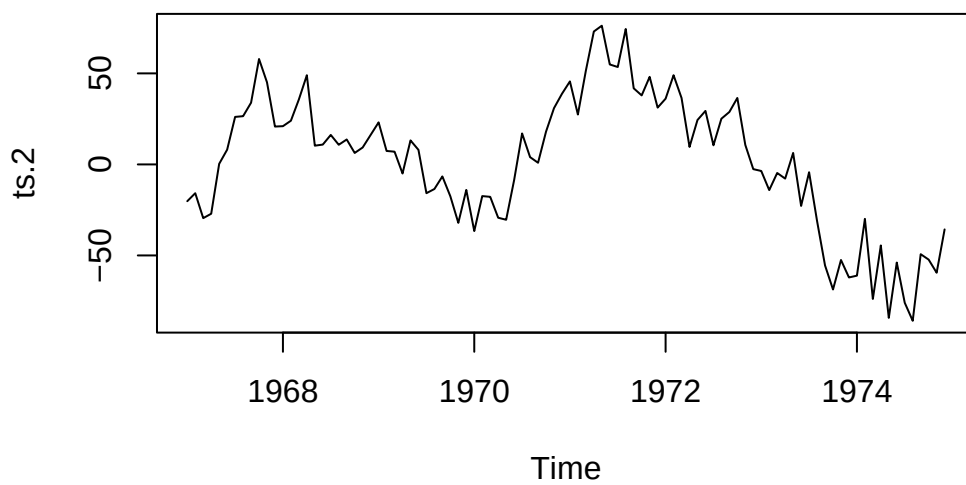


c

```
hstart <- read.table("../data/hstart.dat")  
  
ts <- ts(hstart[, 1], start = c(1966, 1), frequency = 12)  
plot(ts)
```



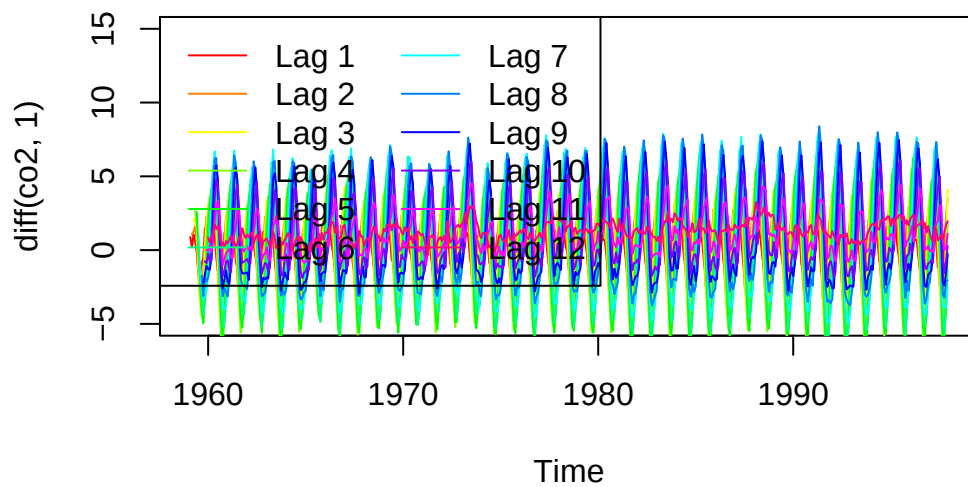
```
ts.2 <- diff(ts, lag=12) # Using lag 12 bc 12 months per year  
plot(ts.2)
```



Exercise 2.3

```
data(co2)
plot(diff(co2, 1), type = "n", ylim = c(-5, 15))

for (l in 1:12) {
  ts_diff <- diff(co2, lag = l)
  lines(ts_diff, col = rainbow(12)[l])
}
legend("topleft", legend = paste("Lag", 1:12), col = rainbow(12), lty = 1, ncol = 2)
```



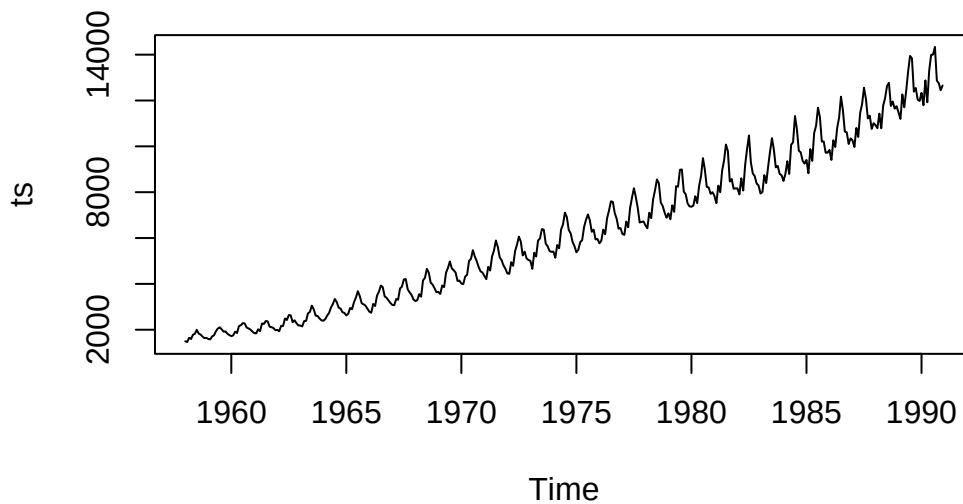
Series 3

Exercise 3.1

- A -> 2
- B -> 3
- C -> 1
- D -> 4

Exercise 3.2

```
cbe <- read.table("./data/cbe.dat", header = TRUE)
ts <- ts(cbe$elec, start = c(1958, 1), frequency = 12)
plot(ts)
```



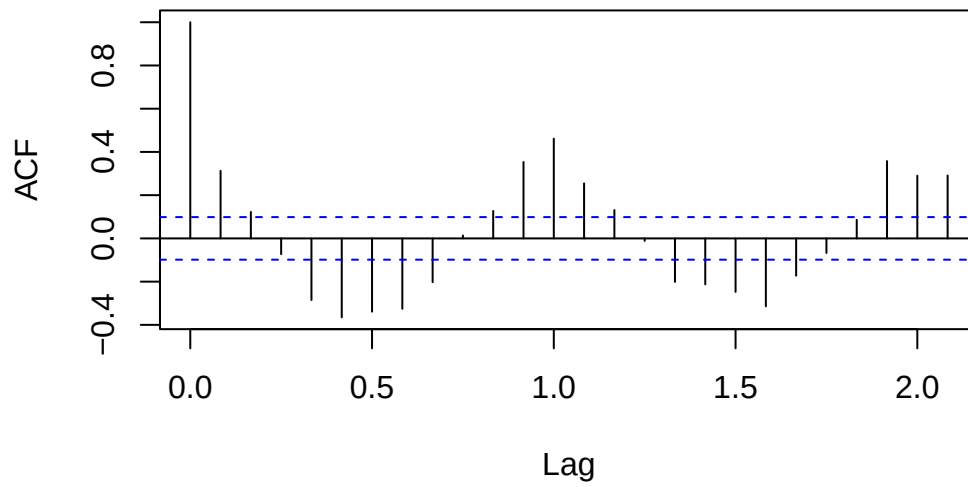
A

At this time, it is not advisable to examine the ACF because the time series is not stationary. We can see trends and seasonality over time. We need to clean that up first before analysing the time series.

B

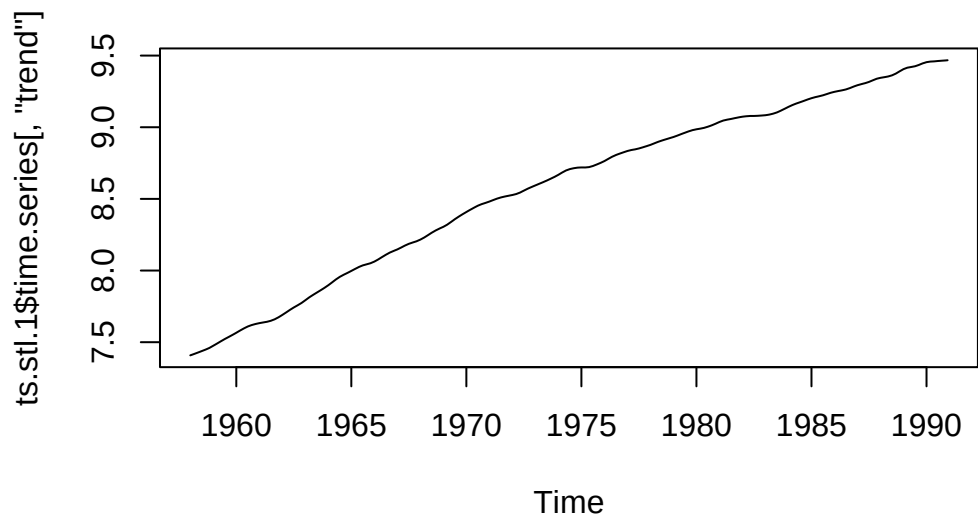
```
decomp <- decompose(ts, type = "multiplicative")
acf(decomp$random, na.action = na.pass)
```

Series decomp\$random



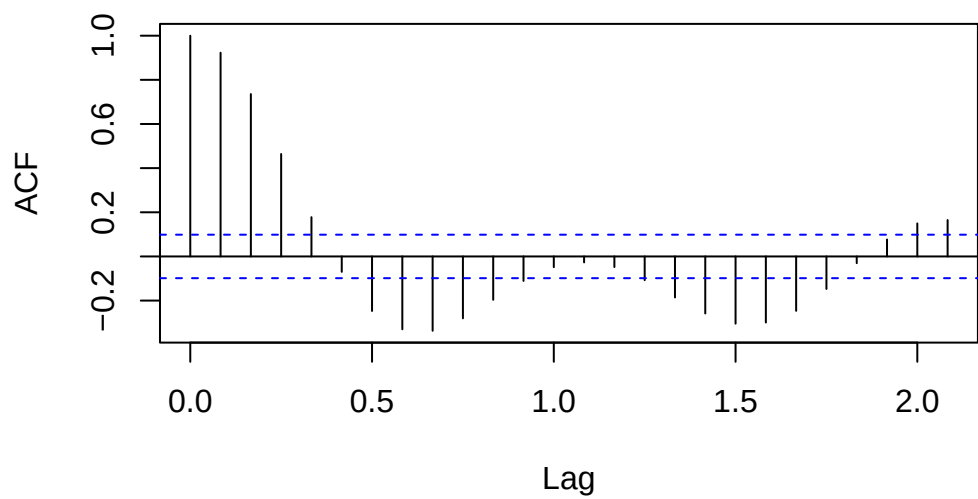
c

```
ts.stl.1 <- stl(log(ts), s.window = "periodic")  
plot(ts.stl.1$time.series[, "trend"])
```

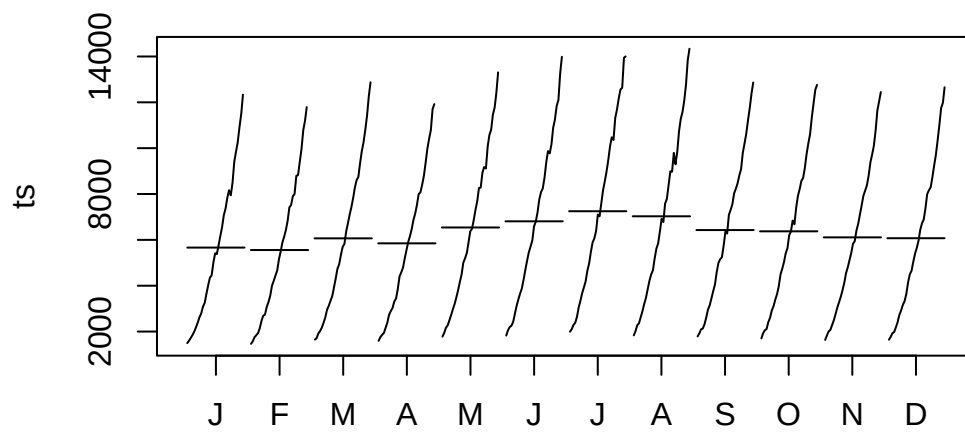


```
decomp <- decompose(ts.stl.1$time.series[, "trend"], type = "multiplicative")  
acf(decomp$random, na.action = na.pass)
```

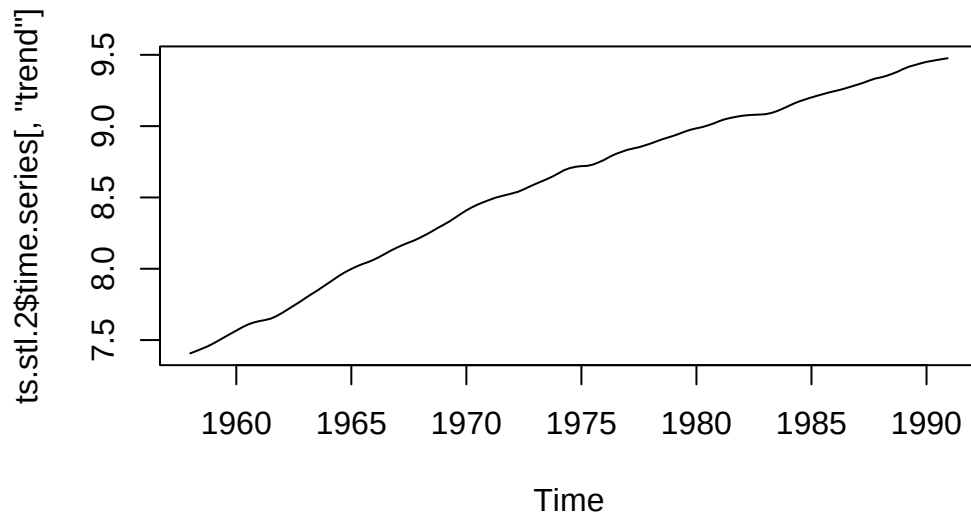
Series decomp\$random



```
monthplot(ts)
```

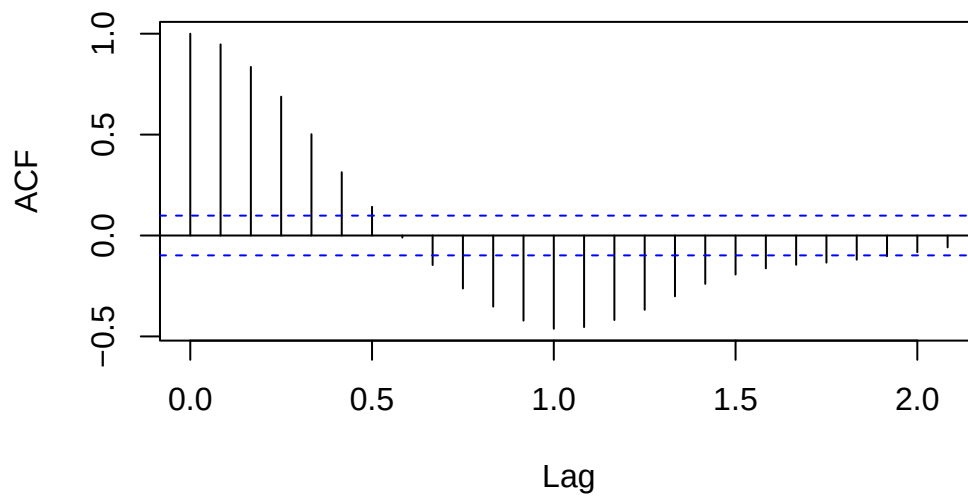


```
ts.stl.2 <- stl(log(ts), s.window = 12)  
plot(ts.stl.2$time.series[, "trend"])
```



```
decomp <- decompose(ts.stl.2$time.series[, "trend"], type = "multiplicative")  
acf(decomp$random, na.action = na.pass)
```

Series decomp\$random



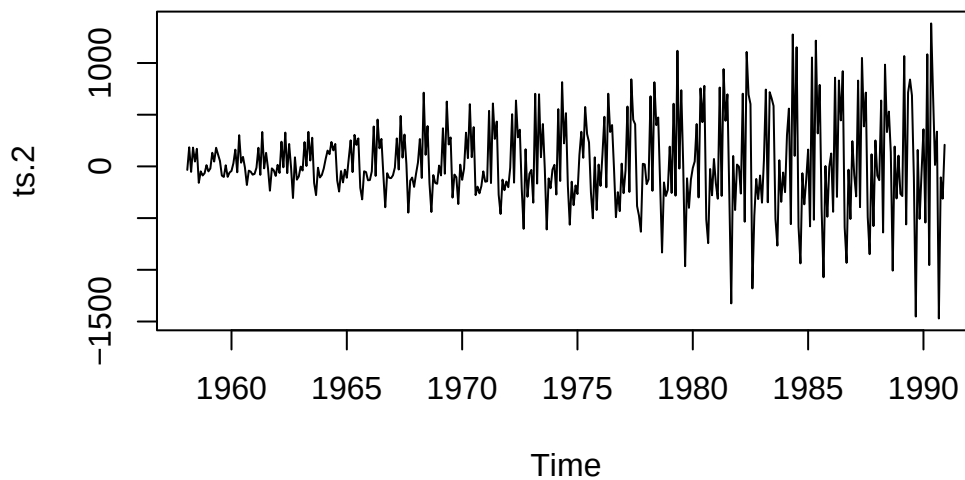
D

1. With `type = "multiplicative"` we assume that seasonality is proportional to the trend.
2. We used `stl(log(ts), ...)` because the original time series had an exponential trend.

E

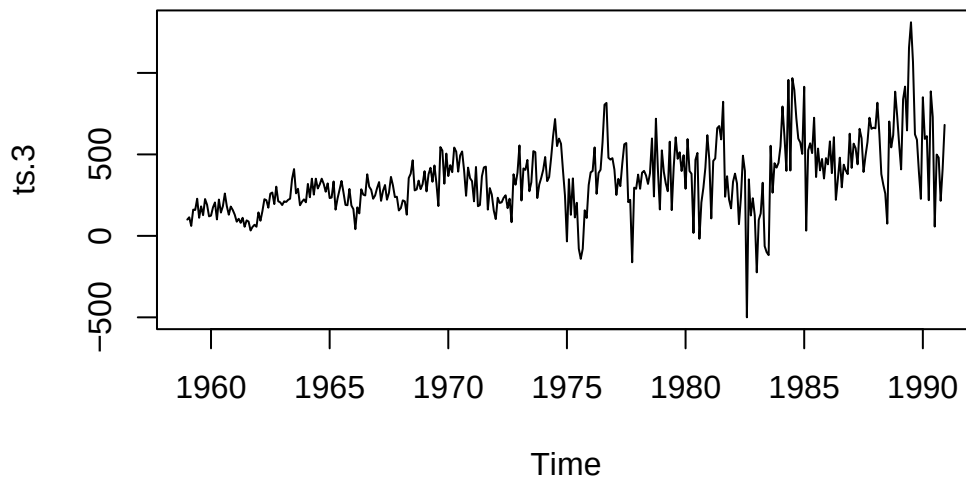
```
cbe <- read.table("./data/cbe.dat", header = TRUE)
ts <- ts(cbe$elec, start = c(1958, 1), frequency = 12)

ts.2 <- diff(ts, lag=1)
plot(ts.2)
```



```
cbe <- read.table("./data/cbe.dat", header = TRUE)
ts <- ts(cbe$elec, start = c(1958, 1), frequency = 12)

ts.3 <- diff(ts, lag=12)
plot(ts.3)
```



Using differencing alone, we eliminate the trend but do not obtain a fully stationary time series.

Series 4

Exercise 4.1

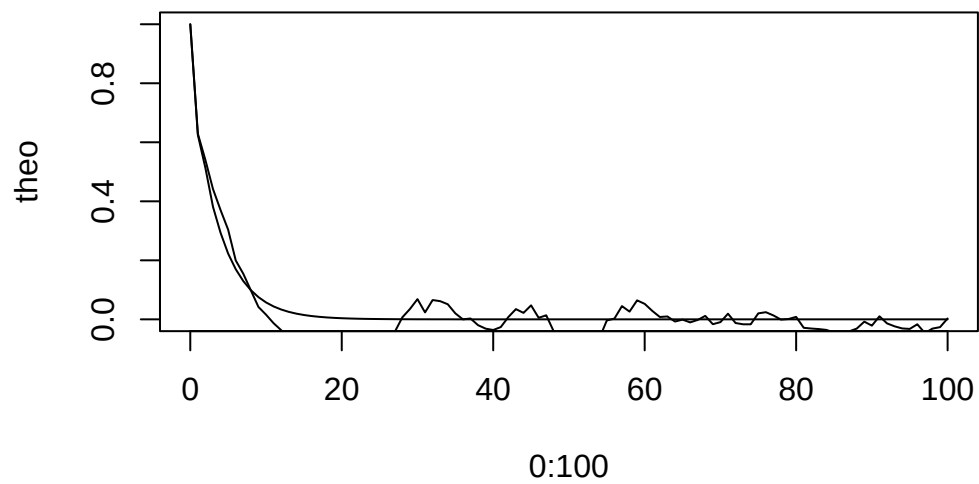
A

```
arima.sim(list(ar = c(0.8)), n = 1000)
```

B

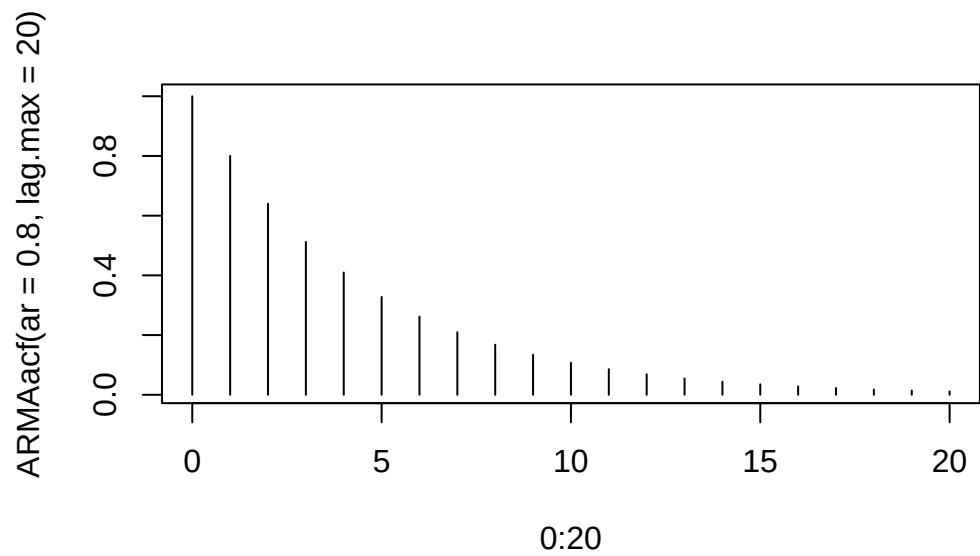
```
theo <- ARMAacf(ar = c(0.5, 0.2), lag.max = 100)
sim_data <- arima.sim(model = list(ar = c(0.5, 0.2)), n = 500)
plugin <- acf(sim_data, lag.max = 100, plot = FALSE)$acf

plot(0:100, theo, type = "l")
lines(0:100, plugin)
```



c

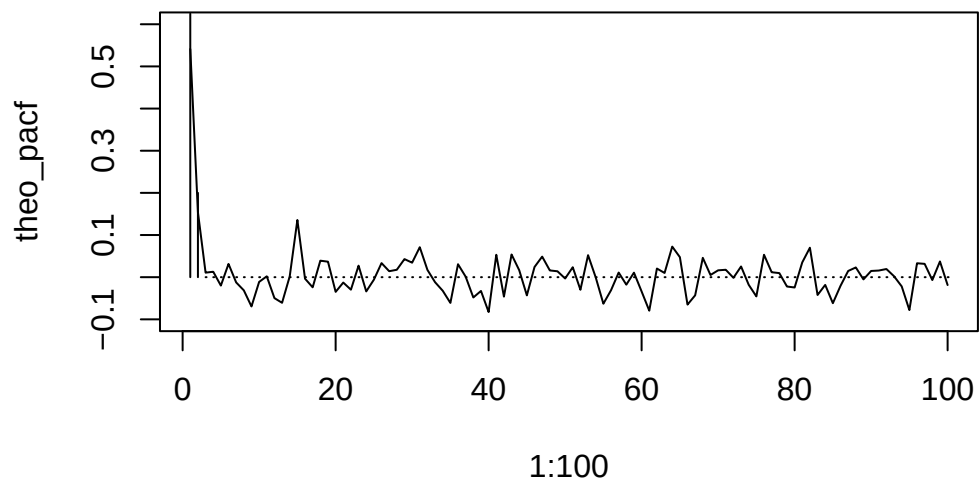
```
plot(0:20, ARMAacf(ar = 0.8, lag.max = 20), type = "h")
```



D

```
theo_pacf <- ARMAacf(ar = c(0.5, 0.2), lag.max = 100, pacf = TRUE)
sim_data <- arima.sim(model = list(ar = c(0.5, 0.2)), n = 500)
estim_pacf <- pacf(sim_data, lag.max = 100, plot = FALSE)$acf

plot(1:100, theo_pacf, type = "h", ylim = c(-0.1, 0.6))
lines(1:100, estim_pacf, type = "l")
```



Exercise 4.2

```
data <- read.table("../data/kreatin.dat", header=TRUE)
```

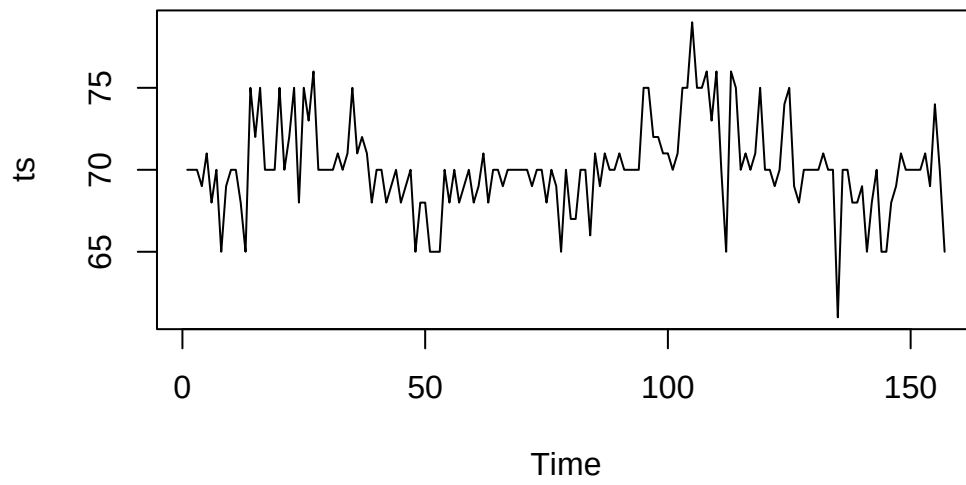
A

The Model should follow a **Random Walk**.

B

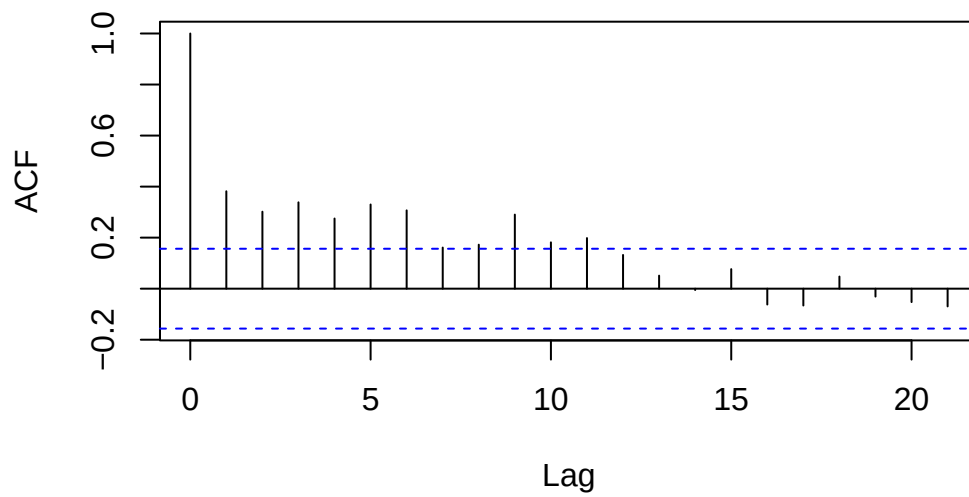
```
ts <- ts(data$gehalt, start=c(1,1))
plot(ts, main="Creatine Concentration of Human Muscular Tissue")
```

Creatine Concentration of Human Muscular Tissue

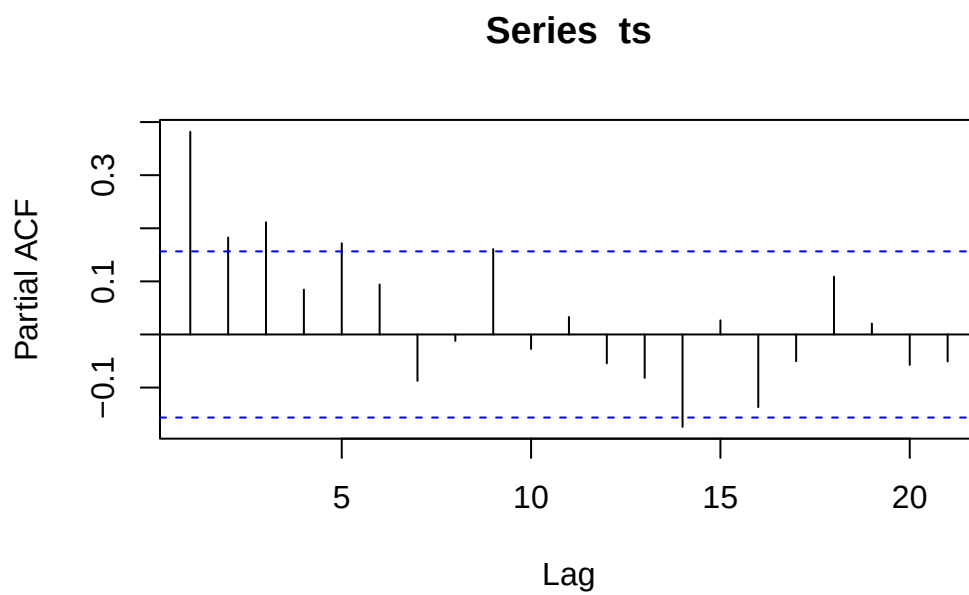


```
acf(ts)
```

Series ts



```
pacf(ts)
```

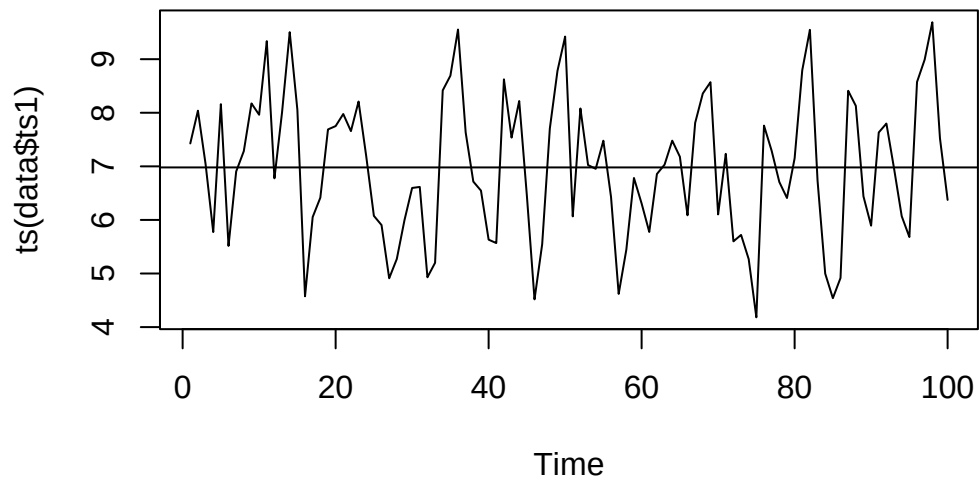


Exercise 4.3

```
data <- read.table("./data/ts_S3_A2.dat", header=TRUE)
```

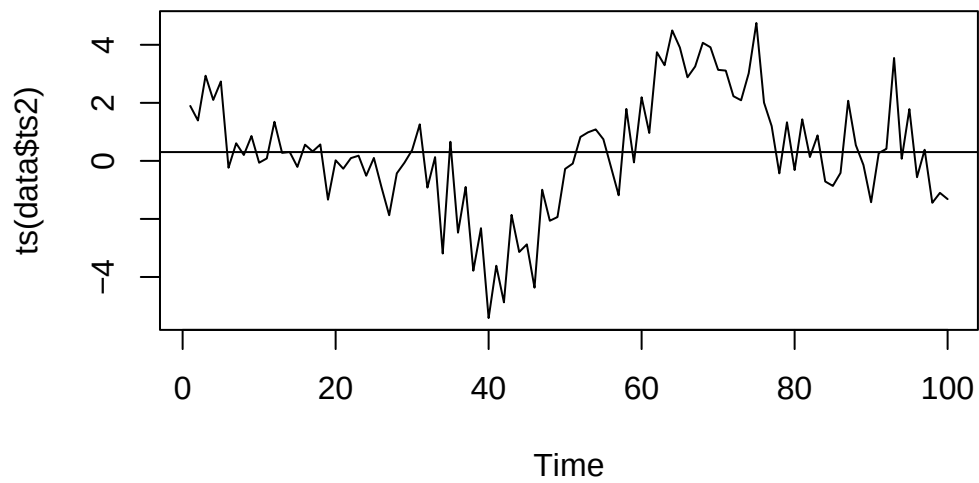
A

```
plot(ts(data$ts1))  
abline(h=mean(data$ts1))
```



`ts1` seems to be stationary.

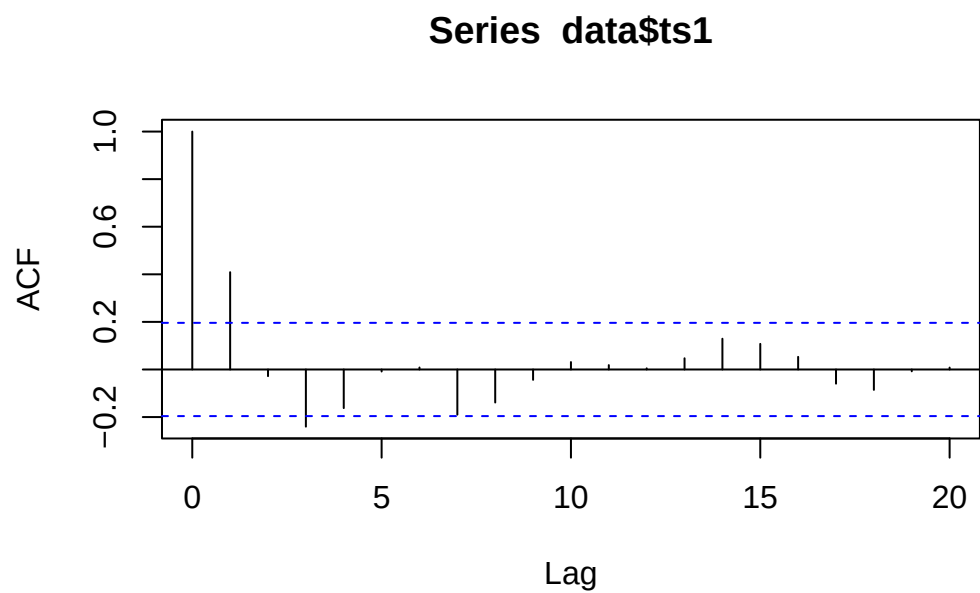
```
plot(ts(data$ts2))  
abline(h=mean(data$ts2))
```



ts2 seems not to be stationary.

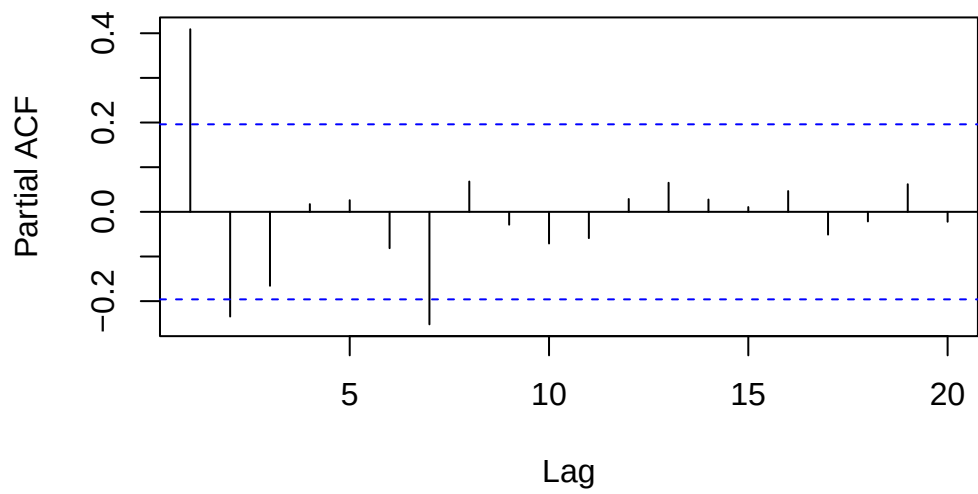
B

```
acf(data$ts1)
```



```
pacf(data$ts1)
```

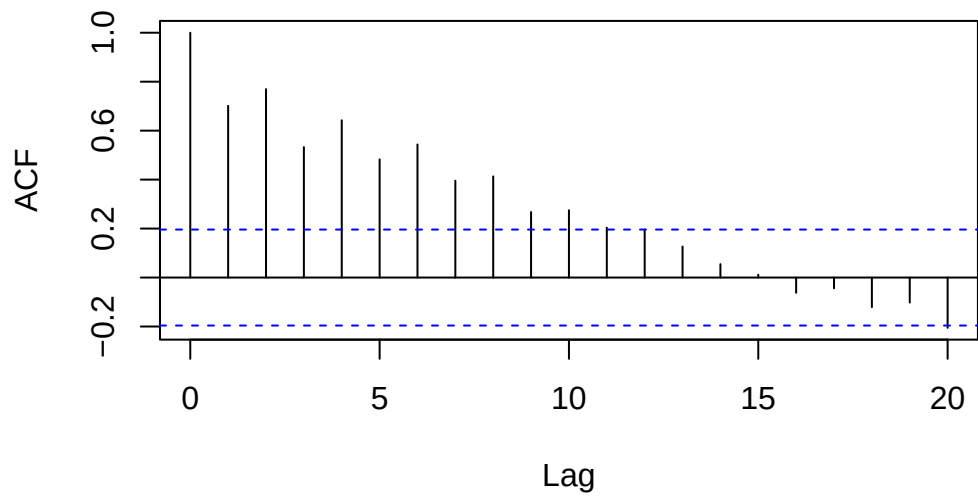
Series data\$ts1



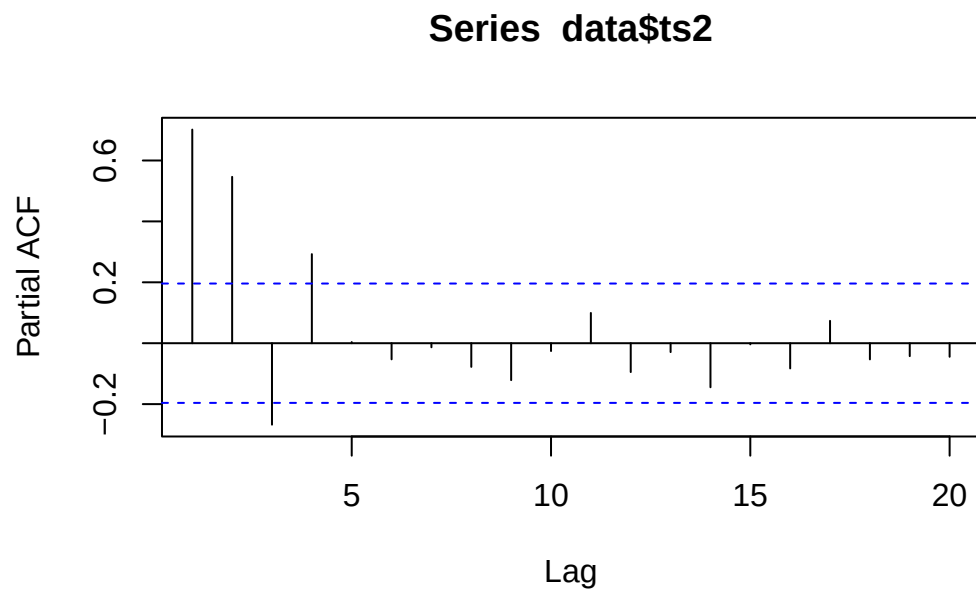
The time series 1 seems to be a $ACF(1)$ -process.

```
acf(data$ts2)
```

Series data\$ts2



```
pacf(data$ts2)
```



The time series 1 seems to be a $ACF(2)$ -process.

Series 5

Task 5.1

A

I

The ACF plot should have a drop at lag 3.

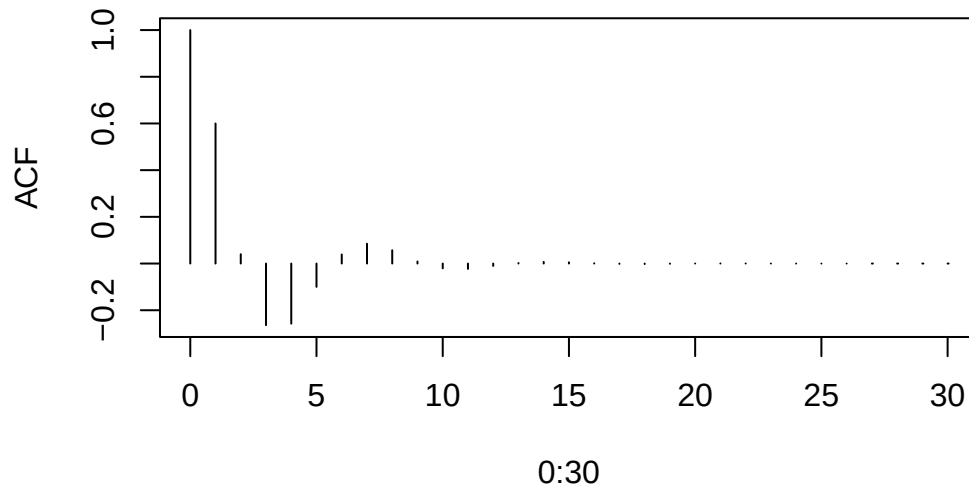
II

```
plot(
  0:30,
  ARMAacf(
    ar = c(0.9, -0.5),
    lag.max = 30
```

```

),
type = "h",
ylab = "ACF"
)

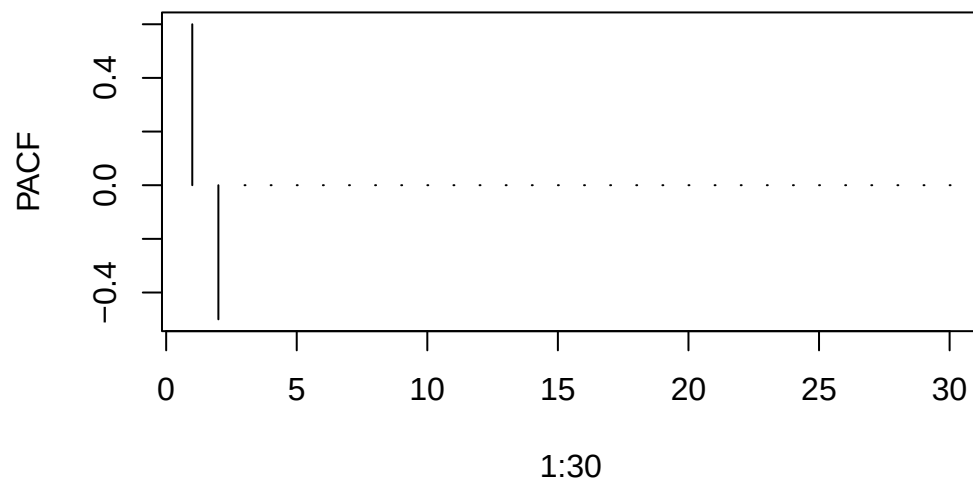
```



```

plot(
  1:30,
  ARMAacf(
    ar = c(0.9, -0.5),
    lag.max = 30,
    pacf = TRUE
  ),
  type = "h",
  ylab = "PACF"
)

```

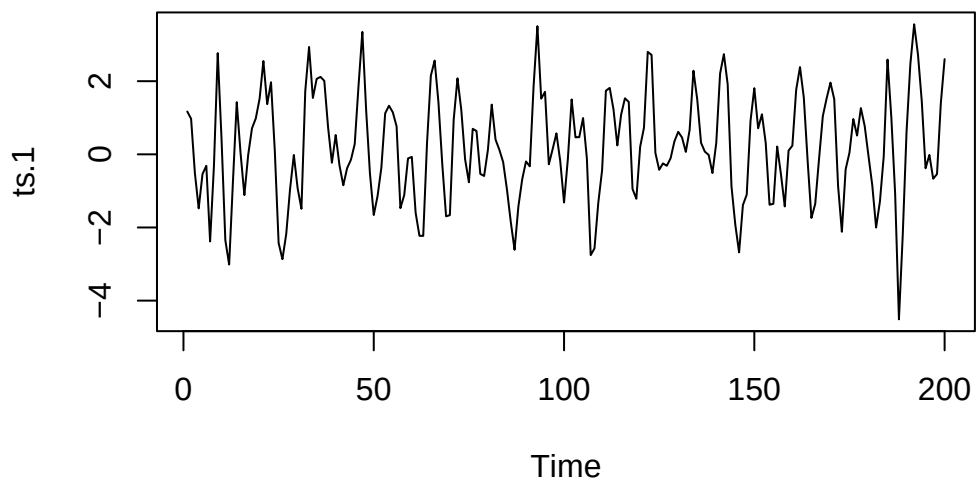


III

```
r.sim1 <- arima.sim(  
  n = 200,  
  model = list(ar = c(0.9, -0.5))  
)
```

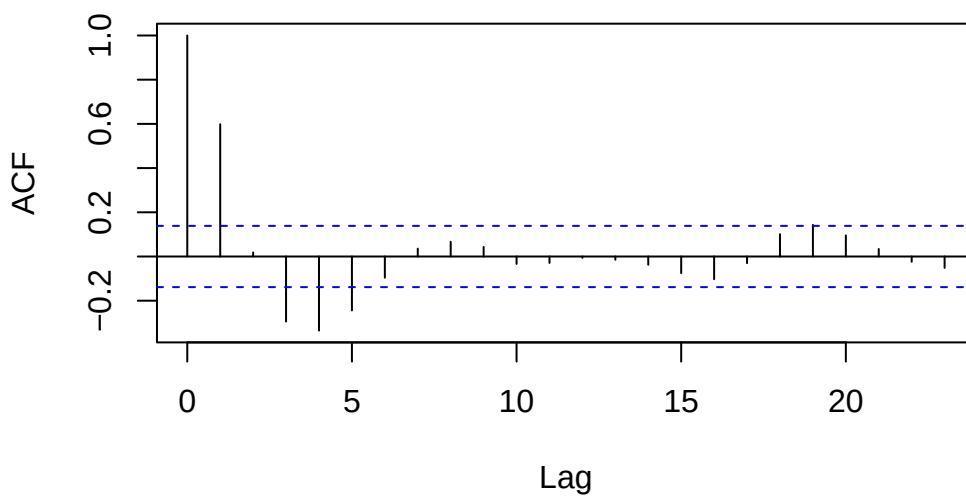
IV

```
ts.1 <- ts(r.sim1)  
plot(ts.1)
```

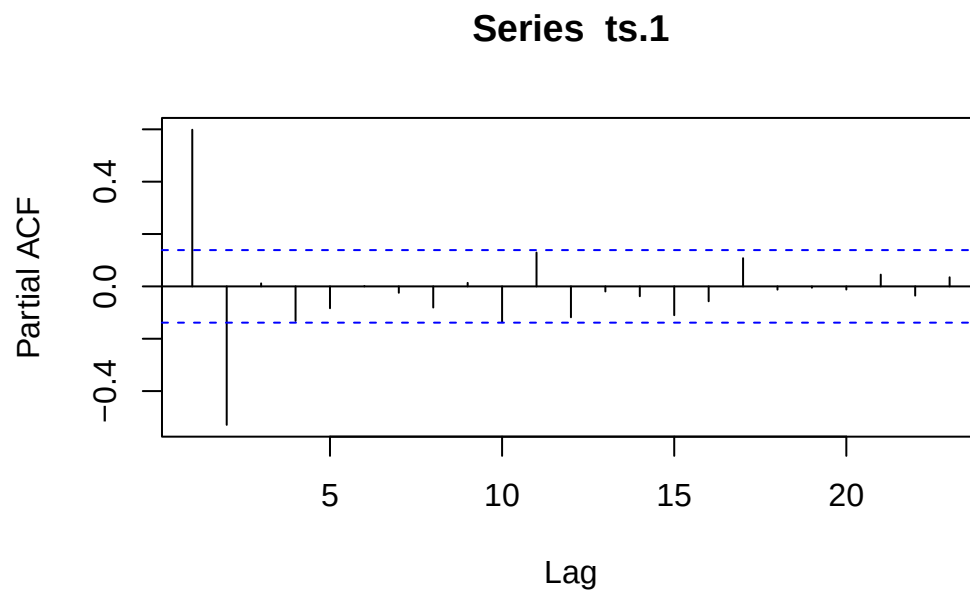


```
acf(ts.1)
```

Series ts.1



```
pacf(ts.1)
```



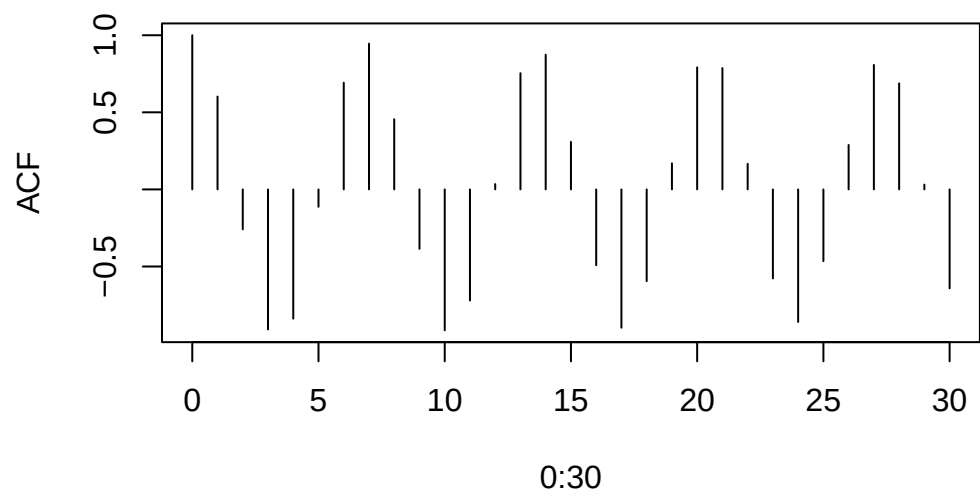
B

I

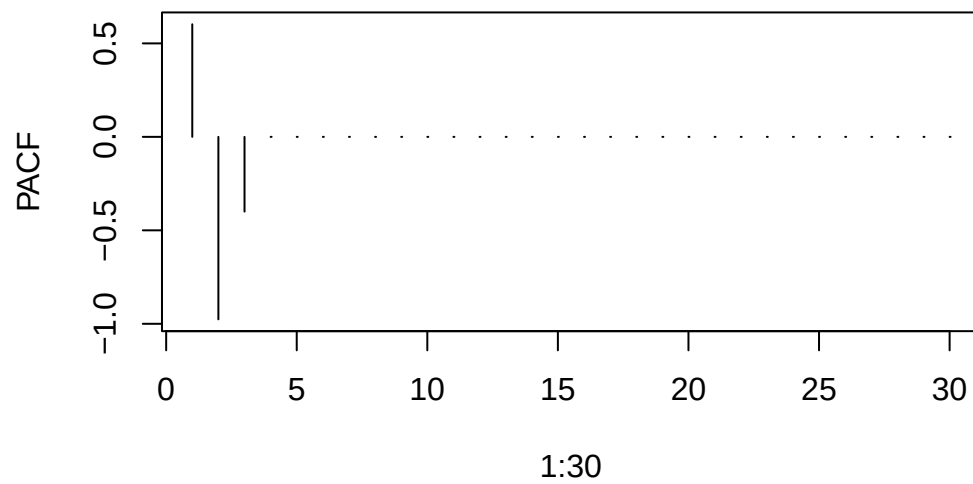
The PACF plot should have a drop at lag 4.

II

```
plot(  
  0:30,  
  ARMAacf(  
    ar = c(0.8, -0.5, -0.4),  
    lag.max = 30  
  ),  
  type = "h",  
  ylab = "ACF"  
)
```



```
plot(
  1:30,
  ARMAacf(
    ar = c(0.8, -0.5, -0.4),
    lag.max = 30,
    pacf = TRUE
  ),
  type = "h",
  ylab = "PACF"
)
```

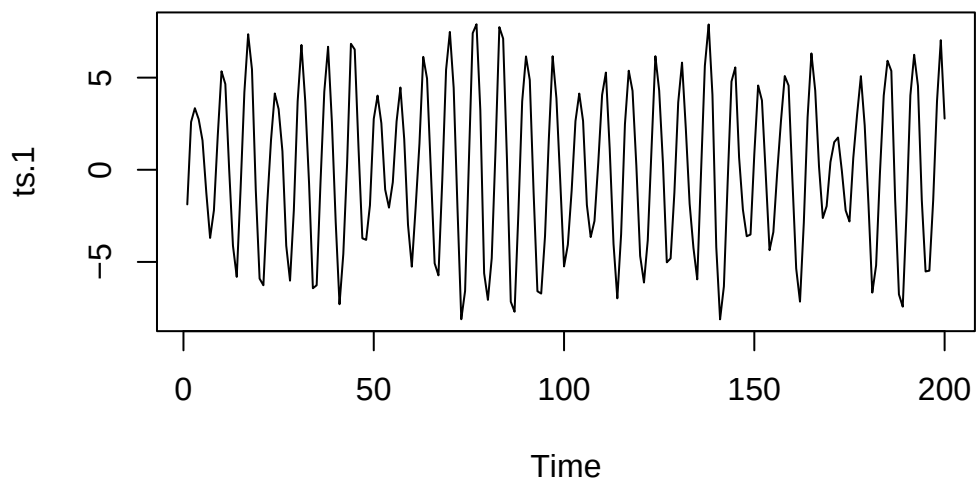


III

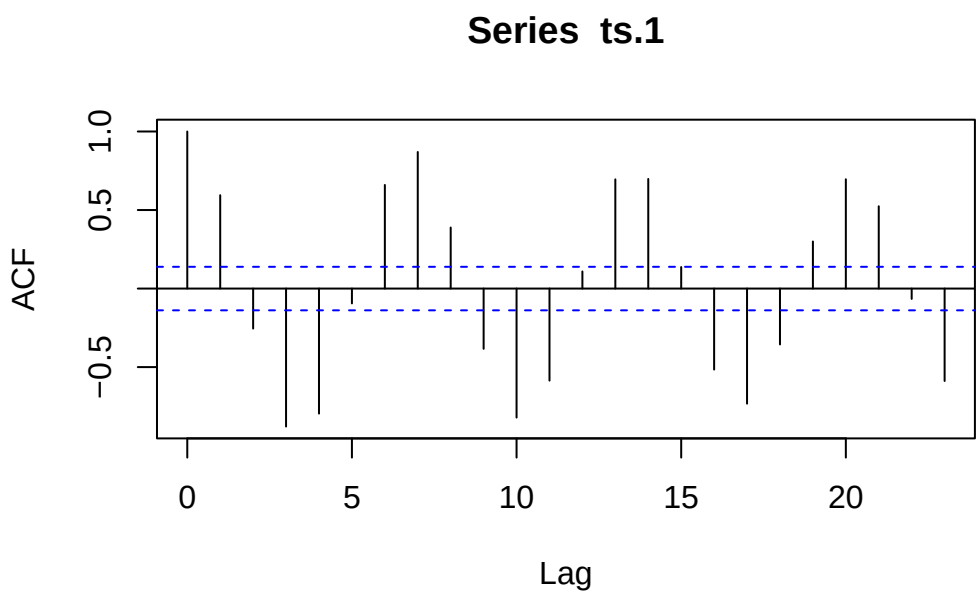
```
r.sim1 <- arima.sim(  
  n = 200,  
  model = list(ar = c(0.8, -0.5, -0.4))  
)
```

IV

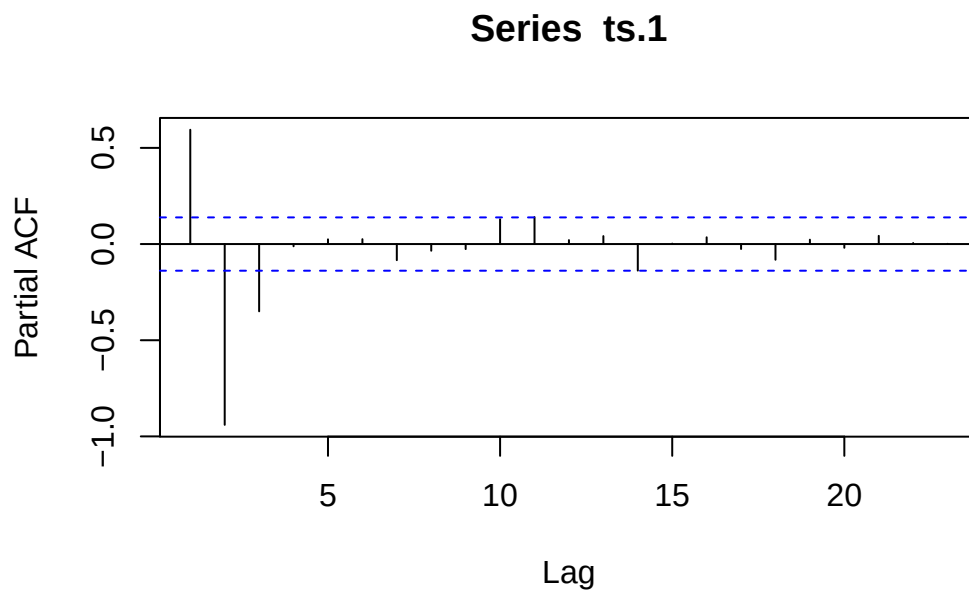
```
ts.1 <- ts(r.sim1)  
plot(ts.1)
```



```
acf(ts.1)
```



```
pacf(ts.1)
```



Exercise 5.2

A

I

```
roots_i <- polyroot(c(1, -0.5, -2))  
abs(roots_i)
```

```
[1] 0.5930703 0.8430703
```

The process is not stationary.

II

```
roots_ii <- polyroot(c(1, -1))  
abs(roots_ii)
```

[1] 1

The process is not stationary.

B

$$0.5 + x = 1$$

$$x = 1 - 0.5$$

$$x = 0.5$$

The process is stationary for $\alpha_2 = 0.5$.

C

The condition for stationarity is $|\alpha| < 1$. Therefore, if $|\alpha| \geq 1$, the root $z = \frac{1}{\alpha}$ lies either on the unit circle ($|z| = 1$) or inside it ($|z| \leq 1$). This mathematically confirms that the model is non-stationary in these cases.

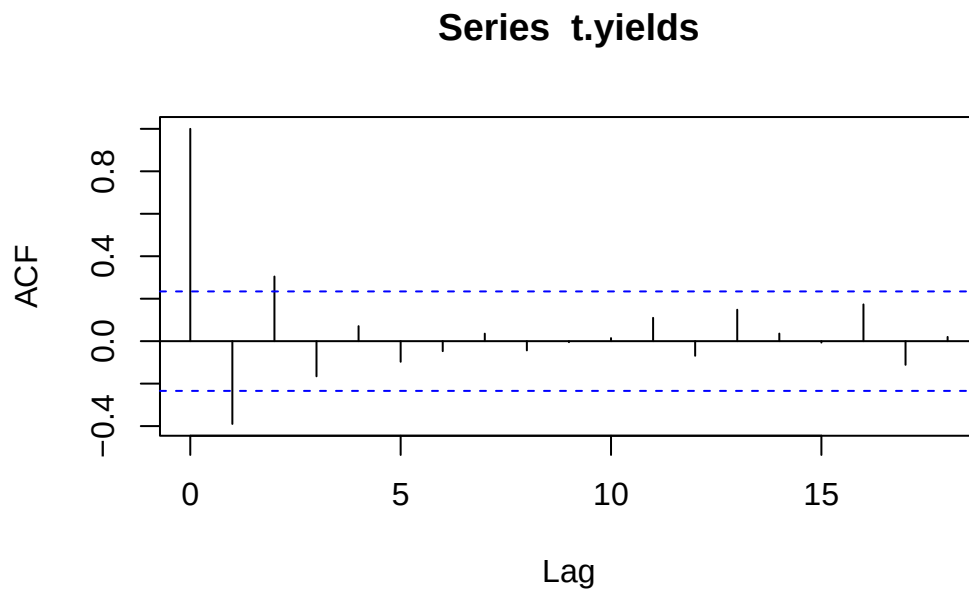
Series 6

Exercise 6.1

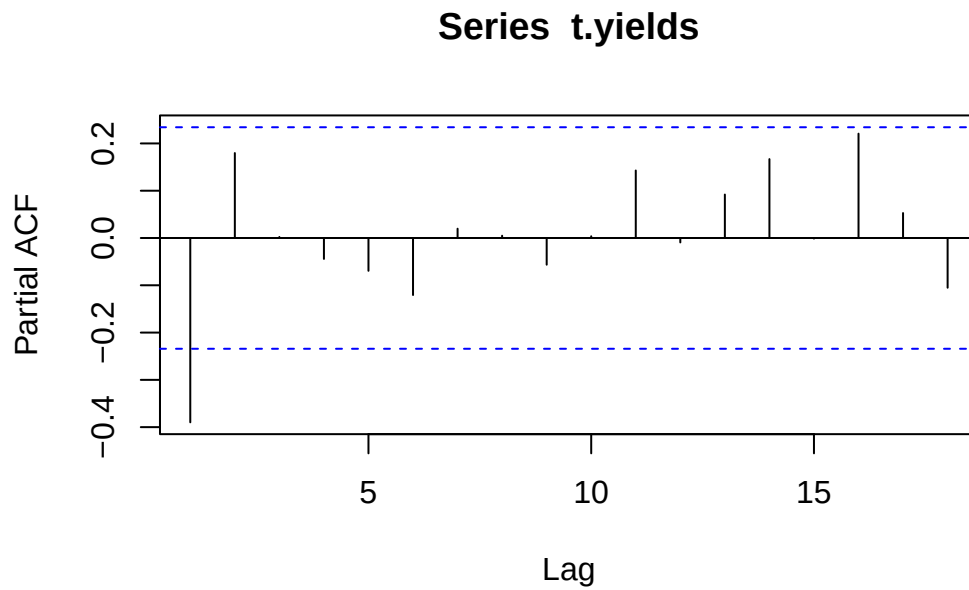
```
yields <- read.table("http://stat.ethz.ch/Teaching/Datasets/WBL/yields.dat", header = FALSE)
t.yields <- ts(yields[, 1])
```

A

```
acf(t.yields)
```



```
pacf(t.yields)
```



The process could be modelled by a $AR(p)$ -Model with the value $p = 3$

B

```
r.yw <- ar(t.yields, method = "yw", order.max = 3)
r.yw$resid
```

Time Series:

Start = 1

End = 70

Frequency = 1

[1]	NA	NA	-23.2701676	8.5624227	-1.7185396	5.1017154
[7]	10.3471780	-11.2034906	3.9364420	1.2089867	17.4563283	-9.2111662
[13]	-2.8577303	-6.3524133	2.2572167	-2.9311234	25.3567725	14.3859910
[19]	-18.0787717	17.6571812	9.7250352	1.7201979	0.7723057	7.4553688
[25]	-3.0885436	2.3171776	1.8505361	-7.5446312	-27.8857718	0.6164494
[31]	6.0842678	18.0959587	11.4294237	20.8583891	5.3106389	2.4003827
[37]	-3.7281741	-0.3234123	-13.1089096	-2.4829289	0.5084360	2.3548536
[43]	-4.3282096	3.2157029	1.8505361	9.4553688	-3.4489132	8.6379522
[49]	-2.7310524	-4.0412731	-11.3891299	3.5161116	8.7481019	-10.3049653
[55]	-2.0635580	0.2900955	-18.1456261	-21.2240340	0.7913572	-2.3118589

```
[61] 14.3954080 -6.6314966 -8.3591694 10.8697651 -9.0885436 2.3982863
[67] -6.4316031 0.8978066 6.7490614 -28.2653704
```

```
str(r.yw)
```

List of 15

```
$ order      : int 2
$ ar         : num [1:2] -0.32 0.18
$ var.pred   : num 120
$ x.mean     : num 51.1
$ aic        : Named num [1:4] 9.84 0.298 0 2
..- attr(*, "names")= chr [1:4] "0" "1" "2" "3"
$ n.used     : int 70
$ n.obs      : int 70
$ order.max  : num 3
$ partialacf : num [1:3, 1, 1] -0.38988 0.17971 0.00226
$ resid      : Time-Series [1:70] from 1 to 70: NA NA -23.27 8.56 -1.72 ...
$ method     : chr "Yule-Walker"
$ series     : chr "t.yields"
$ frequency  : num 1
$ call       : language ar(x = t.yields, order.max = 3, method = "yw")
$ asy.var.coef: num [1:2, 1:2] 0.01444 0.00563 0.00563 0.01444
- attr(*, "class")= chr "ar"
```

Yule-Walker-Koeffizienten:

- $\alpha_1 = -0.32$
- $\alpha_2 = 0.18$

Model:

$$X_t = -0.32 \cdot X_{t-1} + 0.18 \cdot X_{t-2} + \varepsilon_t$$

C

```
r.burg <- ar(t.yields, method = "burg", order.max = 3)
r.burg$resid
```

Time Series:

Start = 1

End = 70

Frequency = 1

```
[1] NA NA -23.0964972 8.1701783 -1.3785617 4.8728473
[7] 10.5505765 -11.1978722 3.7977236 1.3403098 17.3914298 -8.9537665
[13] -3.1238980 -6.2261379 2.0994588 -2.8089304 25.2454443 14.7685021
[19] -18.1260033 17.4687299 10.0560689 1.6430522 0.8457311 7.4219430
[25] -2.9745155 2.2116763 1.9437802 -7.5780570 -27.9582418 0.3118307
[31] 6.2684408 18.0559266 11.6802172 20.8533189 5.5789133 2.3108038
[37] -3.6390123 -0.4223073 -13.0529651 -2.6805103 0.5941087 2.3064707
[43] -4.2697412 3.1267173 1.9437802 9.4219430 -3.3100186 8.5134115
[49] -2.5474727 -4.1848533 -11.3514455 3.3500029 8.8893344 -10.2828313
[55] -2.2022764 0.3468193 -18.1782727 -21.4299662 0.6474050 -2.2228832
[61] 14.3097254 -6.4014872 -8.5781282 10.8990986 -8.9745155 2.2181858
[67] -6.2936738 0.7334424 6.8588212 -28.2490633
```

```
str(r.burg)
```

List of 15

```
$ order      : int 2
$ ar         : num [1:2] -0.332 0.183
$ var.pred   : num 113
$ x.mean     : num 51.1
$ aic        : Named num [1:4] 11.039 0.385 0 1.993
..- attr(*, "names")= chr [1:4] "0" "1" "2" "3"
$ n.used     : int 70
$ n.obs      : int 70
$ order.max  : num 3
$ partialacf : num [1:3, 1, 1] -0.4067 0.183 0.0103
$ resid      : Time-Series [1:70] from 1 to 70: NA NA -23.1 8.17 -1.38 ...
$ method     : chr "Burg"
$ series     : chr "t.yields"
$ frequency  : num 1
$ call       : language ar(x = t.yields, order.max = 3, method = "burg")
$ asy.var.coef: num [1:2, 1:2] 0.0136 0.0053 0.0053 0.0136
- attr(*, "class")= chr "ar"
```

Burg-Koeffizienten:

- $\alpha_1 = -0.332$
- $\alpha_2 = 0.183$

Model:

$$X_t = -0.332 \cdot X_{t-1} + 0.183 \cdot X_{t-2} + \varepsilon_t$$

D

```
r.mle <- ar(t.yields, method = "mle", order.max = 3)
r.mle$resid
```

Time Series:

Start = 1

End = 70

Frequency = 1

[1]	NA	NA	-23.083437837	7.762346509	-1.201980697
[6]	4.561565507	10.602738000	-11.334850093	3.581349256	1.337116357
[11]	17.216926017	-8.885802464	-3.460572312	-6.220191632	1.865930909
[16]	-2.816243379	25.041428463	14.930022474	-18.332496024	17.218498832
[21]	10.197216932	1.428770775	0.782162438	7.273056970	-3.008346873
[26]	2.007435328	1.906279662	-7.726943030	-28.119035966	0.003476264
[31]	6.334762287	17.898351229	11.739491795	20.694392418	5.637512389
[36]	2.088058169	-3.689772086	-0.617858931	-13.115858157	-2.934860904
[41]	0.570317679	2.153690370	-4.337204163	2.944199681	1.906279662
[46]	9.273056970	-3.326921661	8.292016980	-2.521379225	-4.423401367
[51]	-11.422577628	3.121270050	8.899174969	-10.398085740	-2.418650744
[56]	0.292840720	-18.318244153	-21.679521763	0.471483903	-2.242329450
[61]	14.131552552	-6.345881671	-8.876375881	10.832761513	-9.008346873
[66]	1.963159691	-6.288177902	0.491225167	6.843044015	-28.364092605

```
str(r.mle)
```

List of 15

```
$ order      : int 2
$ ar         : num [1:2] -0.341 0.187
$ var.pred   : num 113
$ x.mean     : num 51.2
$ aic        : Named num [1:4] 10.808 0.301 0 1.969
..- attr(*, "names")= chr [1:4] "0" "1" "2" "3"
$ n.used     : int 70
$ n.obs      : int 70
```

```

$ order.max    : num 3
$ partialacf   : NULL
$ resid        : Time-Series [1:70] from 1 to 70: NA NA -23.08 7.76 -1.2 ...
$ method       : chr "MLE"
$ series       : chr "t.yields"
$ frequency    : num 1
$ call         : language ar(x = t.yields, order.max = 3, method = "mle")
$ asy.var.coef: num [1:2, 1:2] 0.0136 0.0053 0.0053 0.0136
- attr(*, "class")= chr "ar"

```

Exercise 6.2

```

d.force <- read.table("http://stat.ethz.ch/Teaching/Datasets/WBL/kraft.dat", header = FALSE)
ts.force <- ts(d.force[, 1])

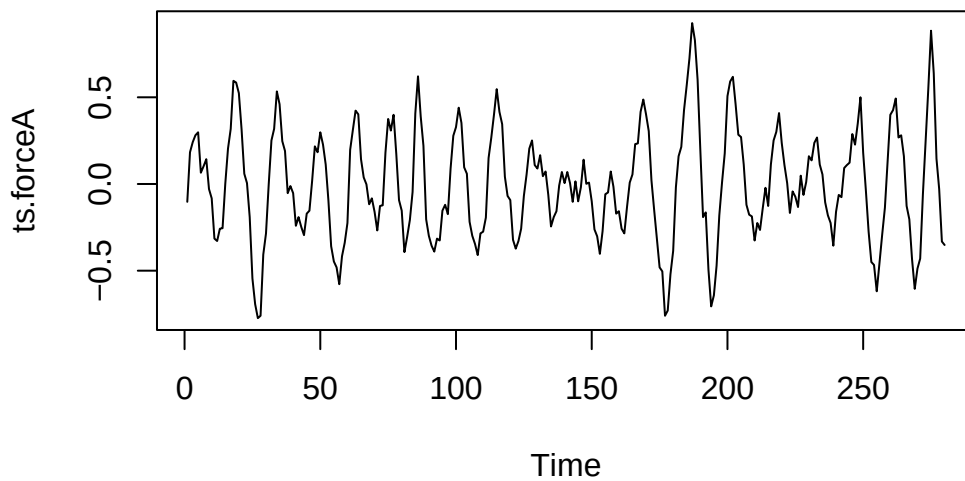
```

A

```

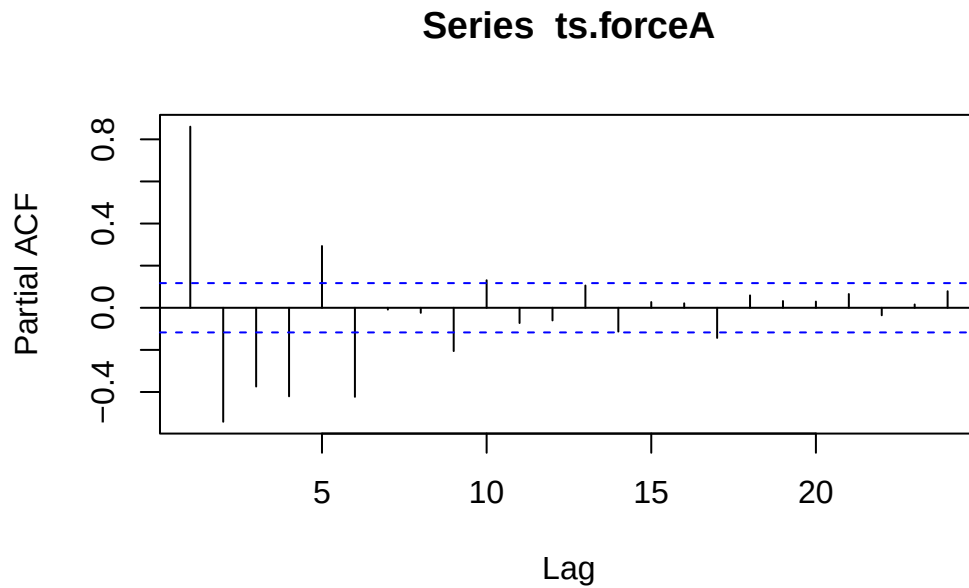
ts.forceA <- window(ts.force, end = 280)
plot(ts.forceA)

```



B

```
pacf(ts.forceA)
```

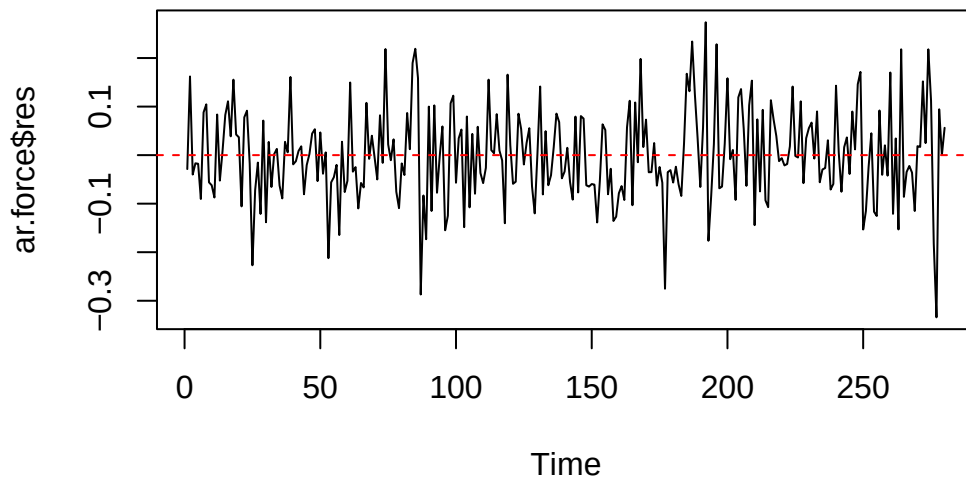


```
ar.force <- ar(ts.forceA, method = "ols", order = 6)
ar.force$aic
```

0	1	2	3	4	5	6
661.05054	283.10981	180.20653	135.93284	80.06312	58.33776	0.00000

C

```
ar.force <- arima(ts.forceA, order = c(6, 0, 0), method = "ML")
plot(ar.force$res)
abline(h=0, col="red", lty=2)
```



The model seems to be appropriate for this time series since the residuals looks normal distributed.

Series 7

Exercise 7.1

A

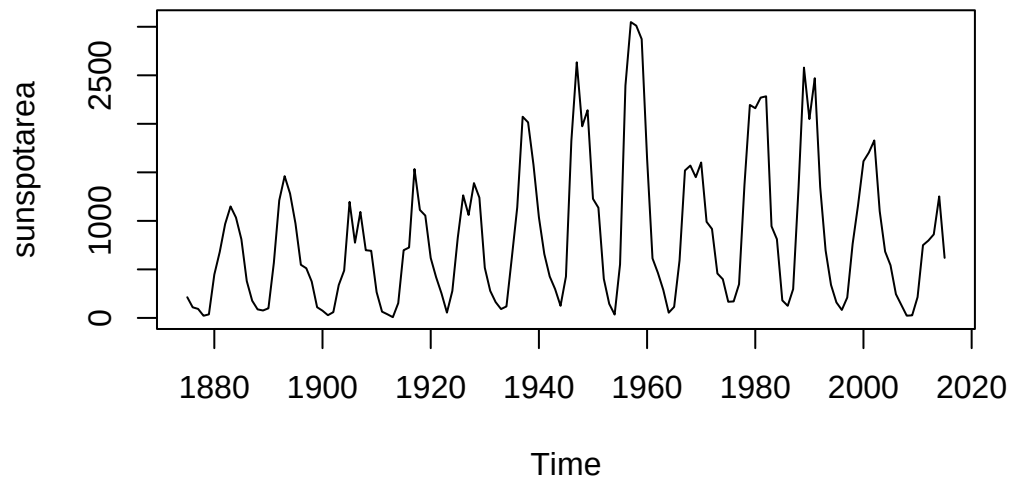
```
arma.1 <-arima.sim(n=250,list(ar=c(-0.75),ma=c(-0.3, 0.25)))
```

B

```
arma.2 <-arima.sim(n=250,list(ar=c(-0.75, -0.3),ma=c(0.25)))
```

Exercise 7.2

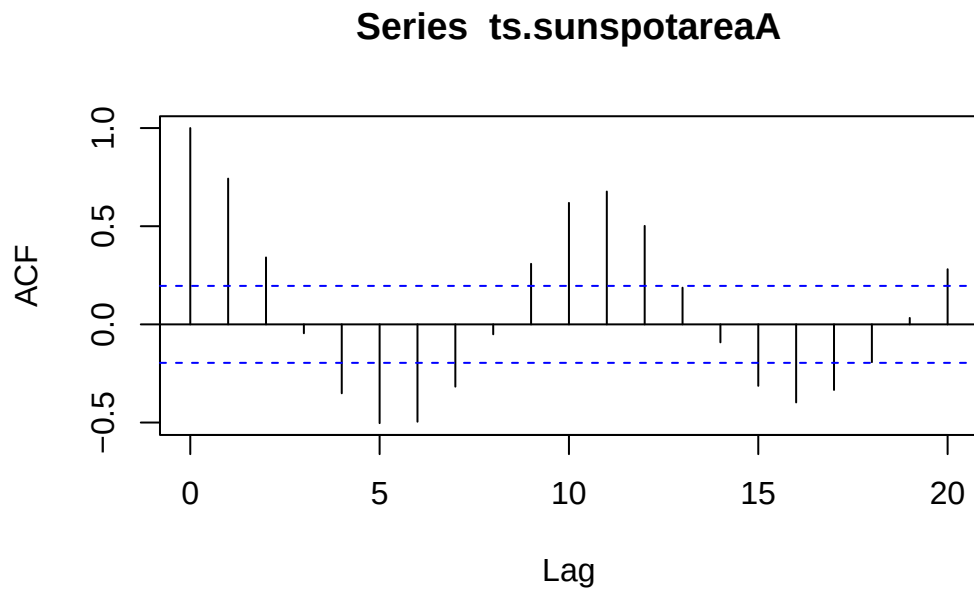
```
library(fpp2)
data(sunspotarea)
plot(sunspotarea)
```



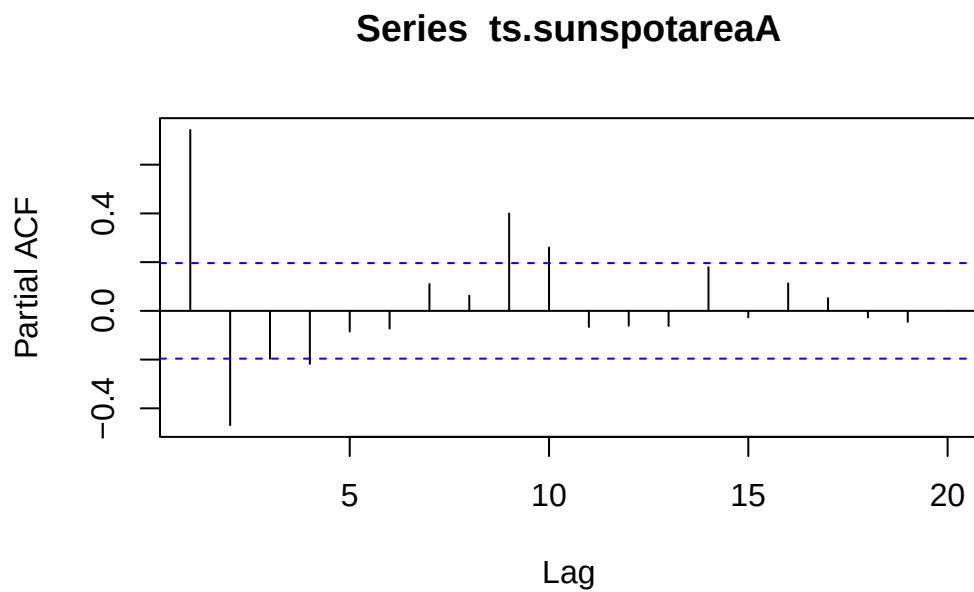
B

```
ts.sunspotarea <- ts(log(sunspotarea))
ts.sunspotareaA <- window(ts.sunspotarea, end = 100)
```

```
acf(ts.sunspotareaA)
```



```
pacf(ts.sunspotareaA)
```



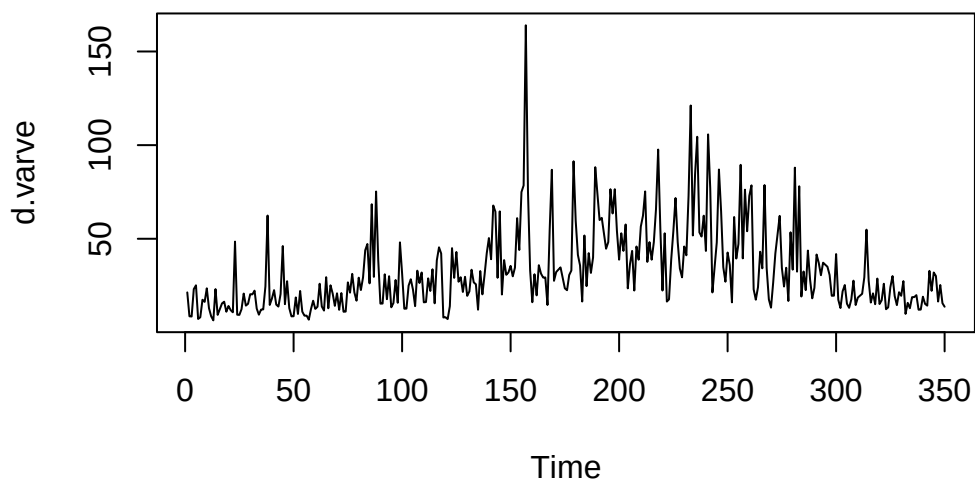
Based on the $(p-)$ acf, we choose a AR -model with the value $p = 3$

Exercise 7.3

```
t.url <- "http://stat.ethz.ch/Teaching/Datasets/WBL/varve.dat"  
d.varve <- ts(scan(t.url)[201:550])
```

A

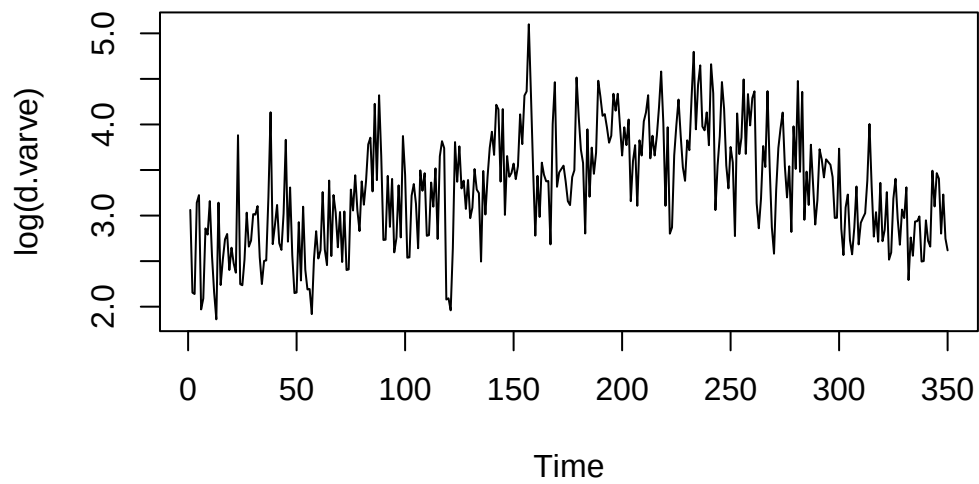
```
plot(d.varve)
```



Yes, there is a need for log-transform.

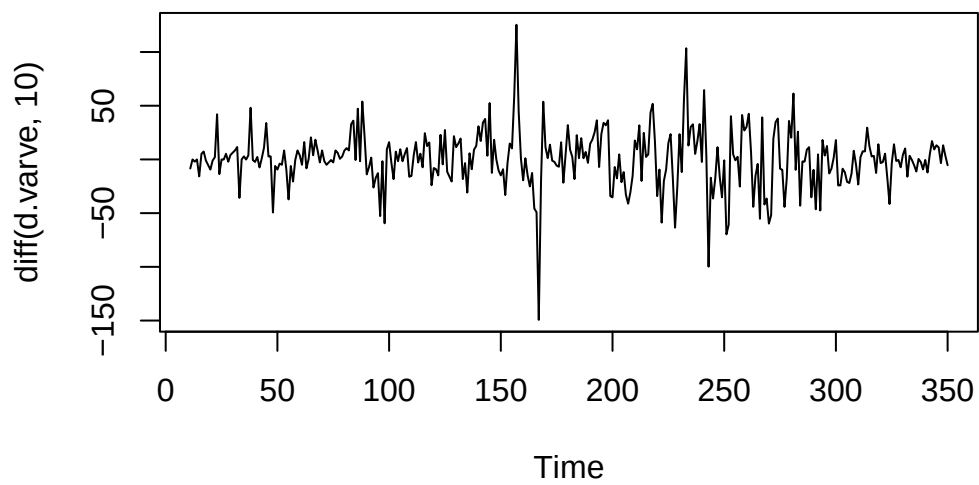
B

```
plot(log(d.varve))
```



The time series is not stationary by simply using log-transformation. We try differencing.

```
plot(diff(d.varve, 10))
```



Looks good now.

Series 8

Exercise 8.1

A

```
arima.1 <-arima.sim(  
  n=250,  
  order=c(2, 1, 2),  
  list(  
    ar=c(0.5, 0,5),ma=c(-0.4, 0.3)  
  )  
)
```

The problem: ar part of model is not stationary!

B

```
# 1. Define the coefficients as polynomials: c(1, -coef1, -coef2)  
ar_poly <- c(1, -0.5, -0.5)  
diff_poly <- c(1, -1)  
  
# 2. Multiply the polynomials using convolution  
# We reverse the second vector (diff_poly) because of how convolve() calculates  
arma_ar_poly <- convolve(ar_poly, rev(diff_poly), type = "open")  
  
# View the resulting polynomial: 1 - 1.5*B + 0*B^2 + 0.5*B^3  
print(arma_ar_poly)
```

```
[1] 1.0 -1.5 0.0 0.5
```

```
# Output: [1] 1.0 -1.5 0.0 0.5
```

```
# 3. Extract the ARMA coefficients  
# Drop the intercept (the first '1') and flip the signs back to standard AR format
```

```
arma_ar_coefs <- -arma_ar_poly[-1]

# 4. The MA coefficients remain identical to the original ARIMA model
arma_ma_coefs <- c(-0.4, 0.3)

# Print final coefficients
cat("Equivalent ARMA AR coefficients:", arma_ar_coefs, "\n")
```

Equivalent ARMA AR coefficients: 1.5 0 -0.5

```
cat("Equivalent ARMA MA coefficients:", arma_ma_coefs, "\n")
```

Equivalent ARMA MA coefficients: -0.4 0.3

Note: Solution generated by AI (Gemini 3.1 Pro).

Exercise 8.2

```
d.beluga <- read.table("http://stat.ethz.ch/Teaching/Datasets/WBL/beluga.dat", header = TRUE)
d.beluga <- ts(d.beluga)
```

A

```
y = lm(NURSING ~ ., data=d.beluga)
y
```

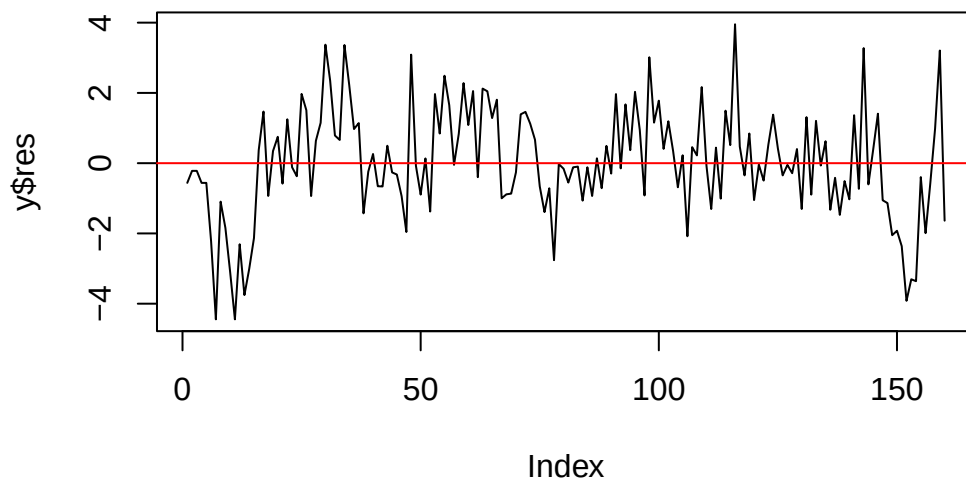
Call:

```
lm(formula = NURSING ~ ., data = d.beluga)
```

Coefficients:

(Intercept)	PERIOD	BOUTS	LOCKONS	DAYNIGHT
0.5602842	0.0001998	0.8784967	2.3903512	-0.3416237

```
plot(y$res, type="l")
abline(h=0, col="red")
```

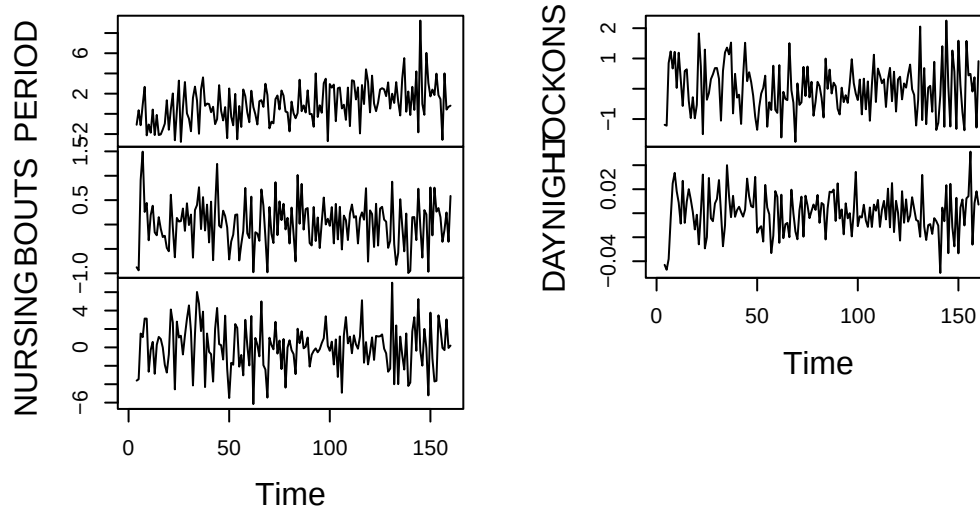


B

```
acf(d.beluga)
```

```
r.burg <- ar(  
  x=d.beluga,  
  order.max=8  
)  
plot(r.burg$res)
```

r.burg\$res



C

```
library(nlme) # Load the package containing the procedure gls()

r.bel.gls <- gls(
  NURSING ~ BOUTS + LOCKONS + DAYNIGHT + PERIOD,
  data = d.beluga,
  correlation = corARMA(
    form = ~PERIOD,
    p = r.burg$order,
    q = 0,
    fixed = FALSE
  ),
  method = "ML"
)

summary(r.bel.gls)
```

Generalized least squares fit by maximum likelihood
 Model: NURSING ~ BOUTS + LOCKONS + DAYNIGHT + PERIOD
 Data: d.beluga

AIC	BIC	logLik
562.0881	589.7647	-272.0441

Correlation Structure: ARMA(3,0)

Formula: ~PERIOD

Parameter estimate(s):

Phi1	Phi2	Phi3
0.25683813	0.35339180	0.04517223

Coefficients:

	Value	Std.Error	t-value	p-value
(Intercept)	1.2919742	0.7810844	1.654078	0.1001
BOUTS	0.2794180	0.3358771	0.831906	0.4067
LOCKONS	2.5827939	0.1966869	13.131499	0.0000
DAYNIGHT	-0.3072369	0.1572643	-1.953634	0.0525
PERIOD	0.0027855	0.0064980	0.428668	0.6688

Correlation:

	(Intr)	BOUTS	LOCKON	DAYNIG
BOUTS	-0.291			
LOCKONS	-0.103	-0.811		
DAYNIGHT	-0.018	-0.132	0.066	
PERIOD	-0.619	-0.228	0.241	0.023

Standardized residuals:

Min	Q1	Med	Q3	Max
-2.80574698	-0.59171135	0.01765555	0.65367271	2.49056057

Residual standard error: 1.576717

Degrees of freedom: 160 total; 155 residual

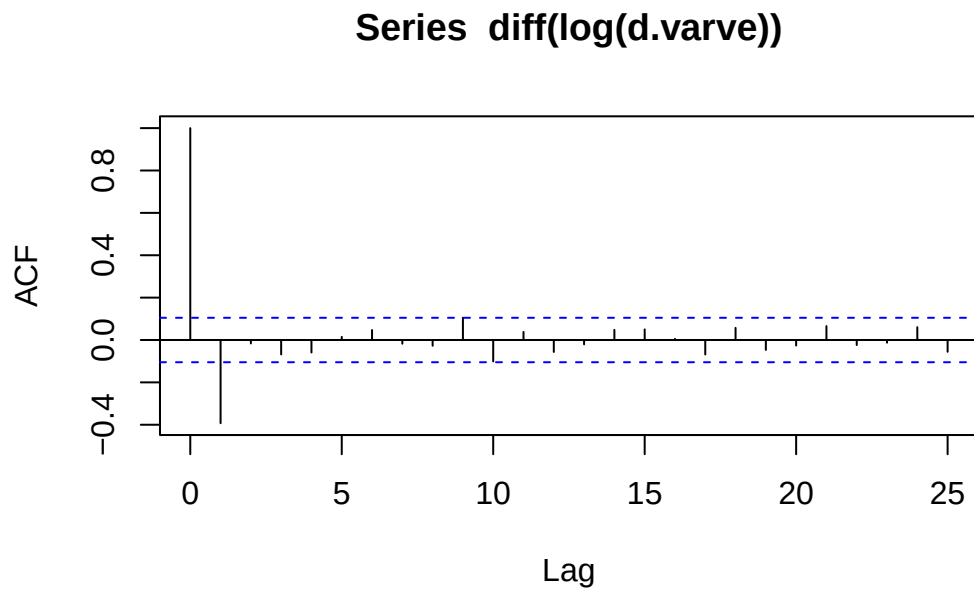
```
d.resid <- ts(resid(r.bel.gls))
```

Exercise 8.3

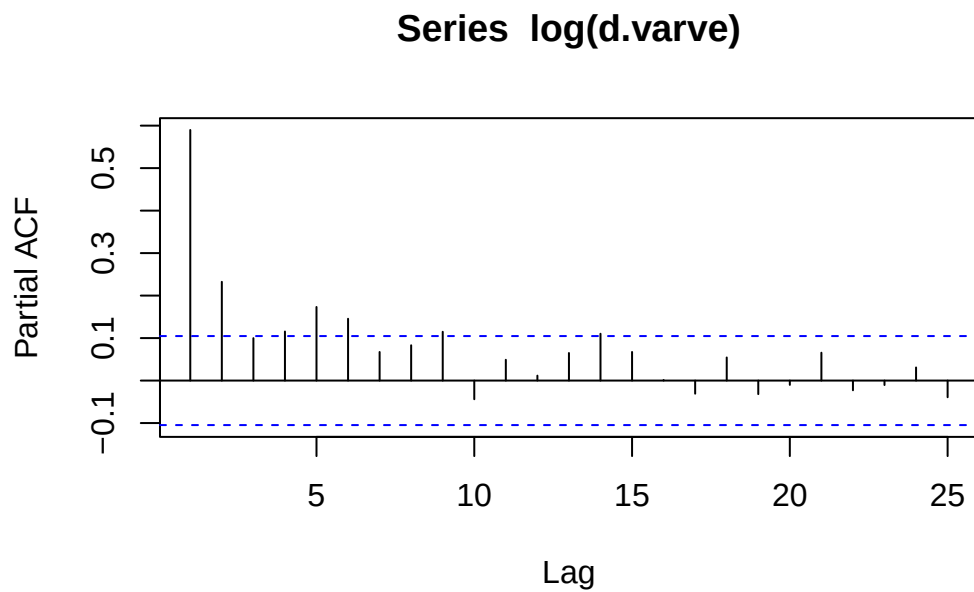
A

```
t.url <- "http://stat.ethz.ch/Teaching/Datasets/WBL/varve.dat"
d.varve <- ts(scan(t.url)[201:550])
```

```
acf(diff(log(d.varve)))
```



```
pacf(log(d.varve))
```



```
r.varve <- arima(
  x=log(d.varve),
  order = c(2, 1, 2),
  method="ML"
)
r.varve
```

Call:

```
arima(x = log(d.varve), order = c(2, 1, 2), method = "ML")
```

Coefficients:

	ar1	ar2	ma1	ma2
	-0.1442	0.1452	-0.5334	-0.3548
s.e.	0.5439	0.1345	0.5453	0.4968

sigma^2 estimated as 0.2132: log likelihood = -226.21, aic = 462.42

B

$$y = -0.1442\alpha_1 + 0.1452\alpha_2 + -0.5334\beta_1 + -0.3548\beta_2 + \varepsilon_0$$

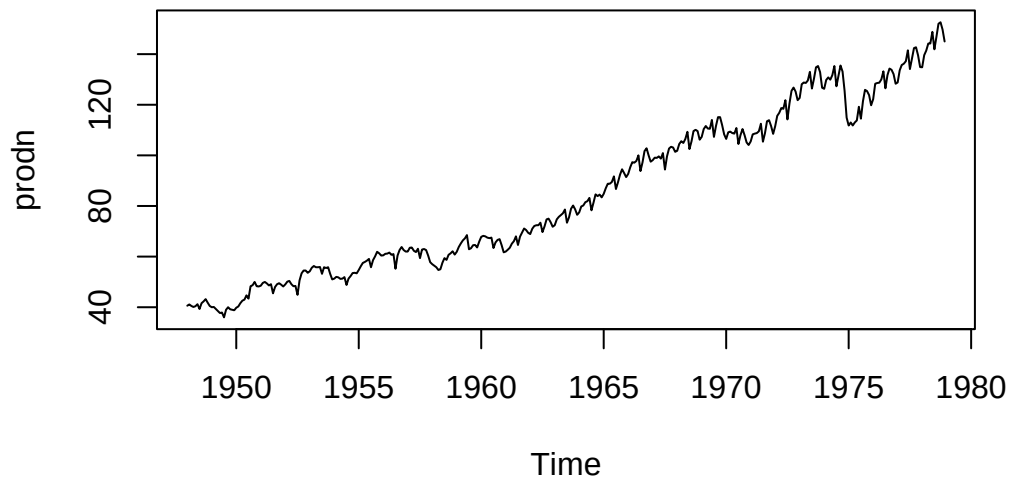
Series 9

Exercise 9.1

A

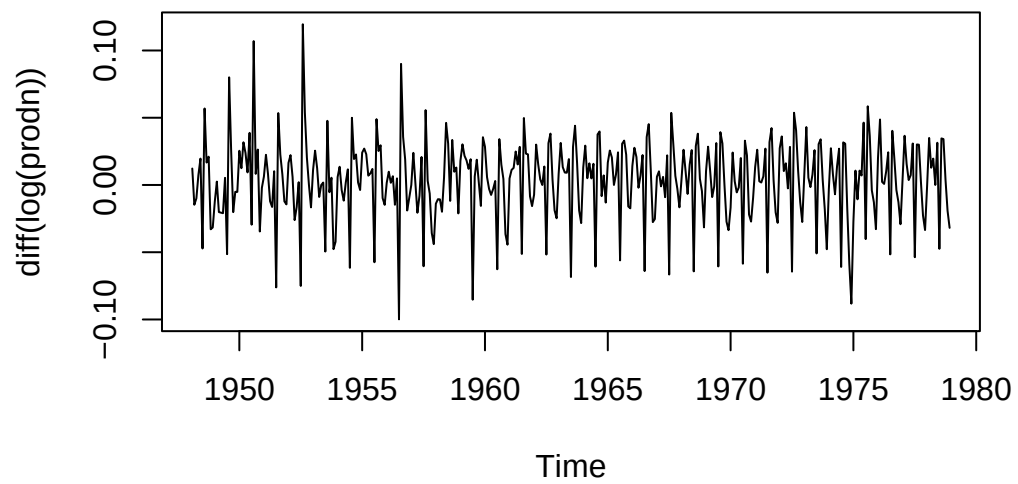
```
library(astsa)
data(prodn)

plot(prodn)
```



B

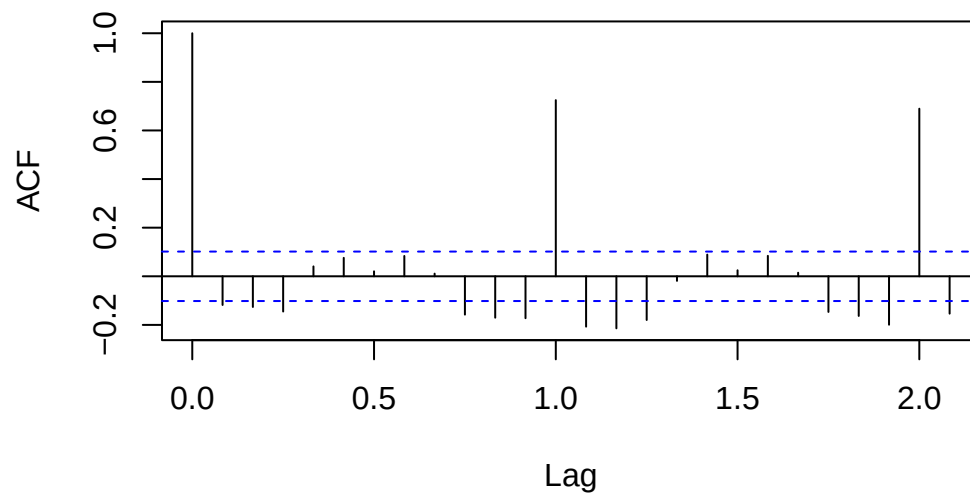
```
plot(diff(log(prodn)))
```



c

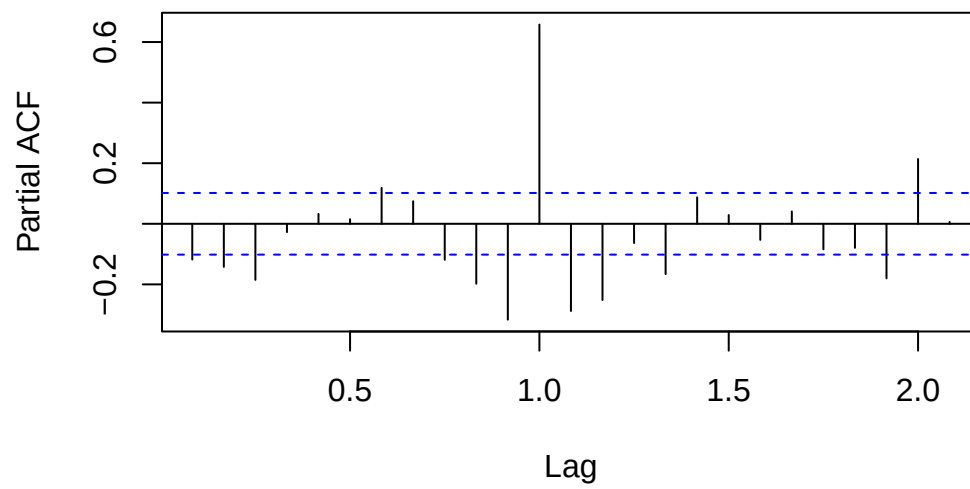
```
prodn.trans <- diff(log(prodn))  
acf(prodn.trans)
```

Series prodn.trans



```
pacf(prodn.trans)
```

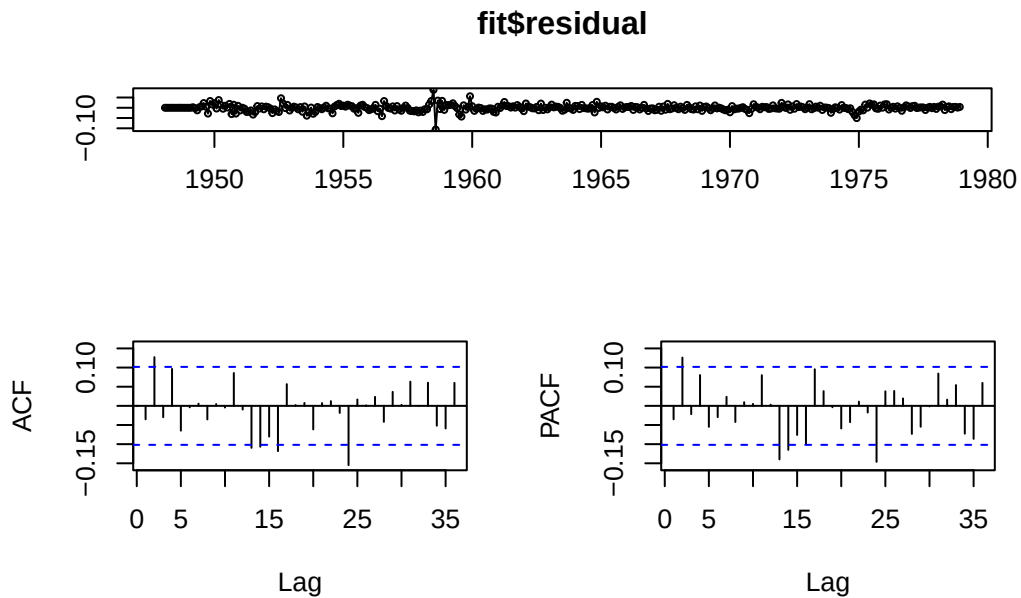
Series prodn.trans



```
library(forecast)

fit<-arima(
  x=prodn.trans,
  order=c(1,1,1),
  seasonal=c(1,1,1)
)

tsdisplay(fit$residual)
```



```
fitAuto<-auto.arima(prodn.trans, ic="aic")
fitAuto
```

Series: prodn.trans
ARIMA(3,0,0)(2,1,1)[12]

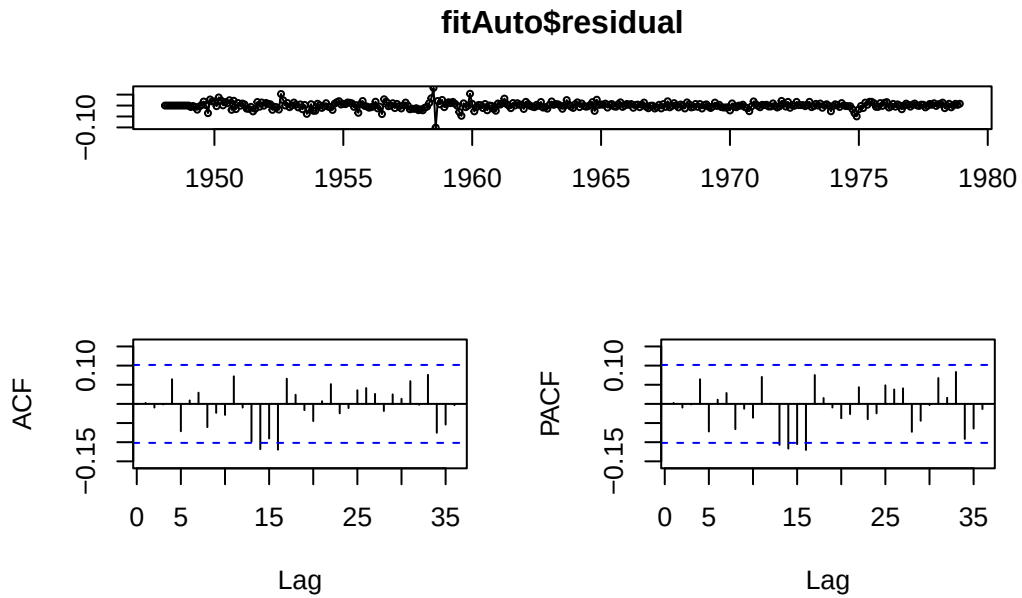
Coefficients:

	ar1	ar2	ar3	sar1	sar2	sma1
	0.2431	0.1458	-0.0493	-0.1832	-0.240	-0.6940
s.e.	0.0527	0.0544	0.0531	0.0708	0.063	0.0583

$\sigma^2 = 0.0002249$: log likelihood = 993.91

AIC=-1973.83 AICc=-1973.51 BIC=-1946.65

```
tsdisplay(fitAuto$residual)
```



Series 10

Exercise 10.1

```
library(fpp2)

lsunspot100 <- window(log(sunspotarea), start = 1875, end = 1974)
fit.ar10 <- arima(lsunspot100, order = c(10, 0, 0))
fit.ar10
```

Call:

```
arima(x = lsunspot100, order = c(10, 0, 0))
```

Coefficients:

	ar1	ar2	ar3	ar4	ar5	ar6	ar7	ar8
	0.7171	-0.1887	0.0045	-0.0621	0.0220	-0.1140	0.1694	-0.2284
s.e.	0.0938	0.1175	0.1176	0.1180	0.1159	0.1183	0.1184	0.1189

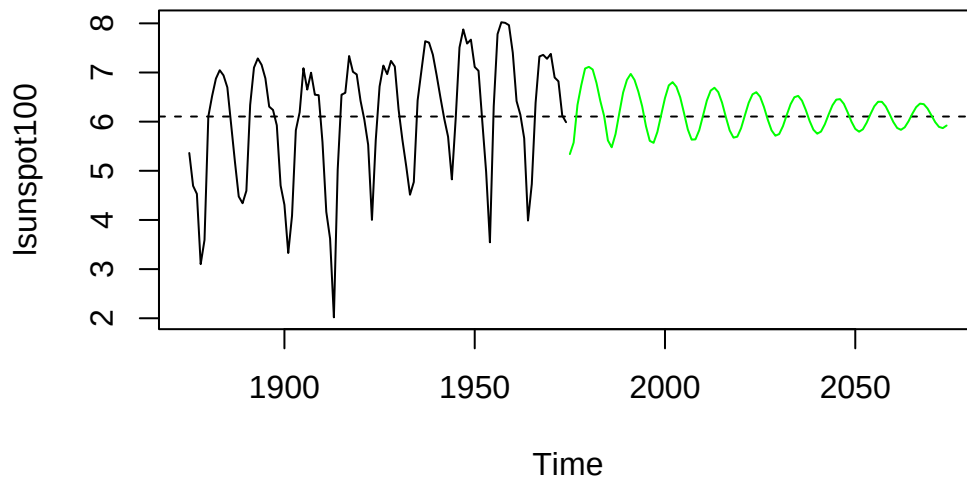
	ar9	ar10	intercept
	0.1272	0.3601	6.1045
s.e.	0.1207	0.0970	0.2597

sigma^2 estimated as 0.3439: log likelihood = -91.1, aic = 206.2

A

```
pred <- predict(fit.ar10, n.ahead = 100)

plot(x=lsunspot100, xlim = c(1875, 2074))
abline(h = fit.ar10$coef["intercept"], lty = 2)
lines(pred$pred, col = "green")
```



B

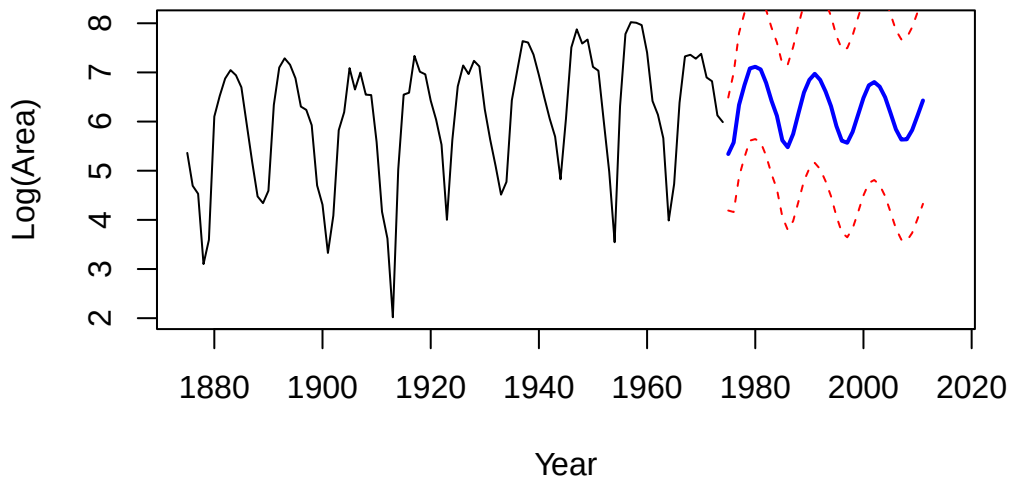
```

pred_b <- predict(fit.ar10, n.ahead = 37)

plot(lsunspot100,
     main = "Log-Transformed Sunspot Area with AR(10) Predictions",
     ylab = "Log(Area)",
     xlab = "Year",
     xlim = c(1875, 2015))
lines(pred_b$pred, col = "blue", lwd = 2)
lines(pred_b$pred + 1.96 * pred_b$se, col = "red", lty = 2)
lines(pred_b$pred - 1.96 * pred_b$se, col = "red", lty = 2)

```

Log-Transformed Sunspot Area with AR(10) Predictions



```

# 6. Calculate the Mean Squared Forecasting Error (MSFE)
# Extract the actual observed values for the out-of-sample period (1975-2011)
actual_out_of_sample <- window(sunspotarea, start = 1975, end = 2011)

# Compute the Mean Squared Error between actuals and predictions
msfe <- mean((actual_out_of_sample - pred_b$pred)^2)

cat("Mean Squared Forecasting Error (MSFE):", msfe, "\n")

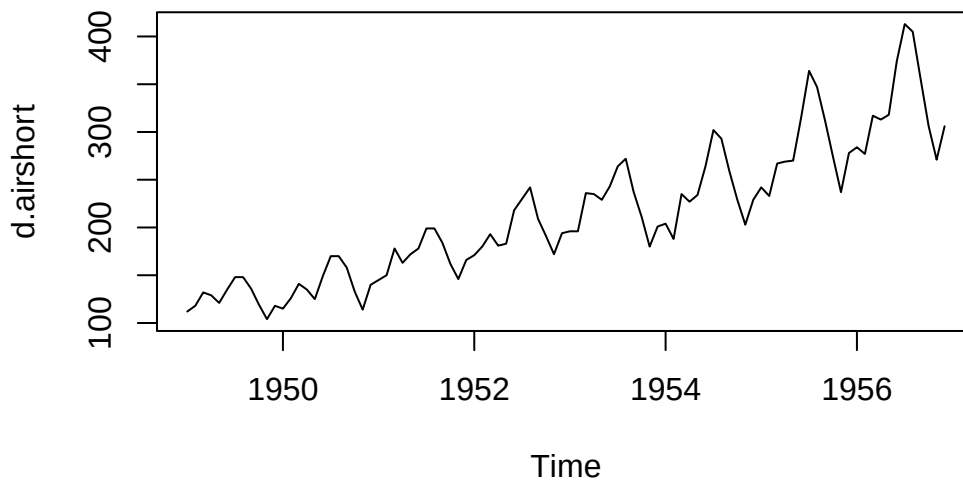
```

Mean Squared Forecasting Error (MSFE): 1562725

Note: Solution generated by AI (Gemini 3.1 Pro).

Exercise 10.2

```
data("AirPassengers")
d.air <- AirPassengers
d.airshort <- window(d.air, end = c(1956, 12))
plot(d.airshort)
```



A

```
d.airshort.trans <- diff(log(d.airshort))
fit <- auto.arima(d.airshort.trans, ic="aic")

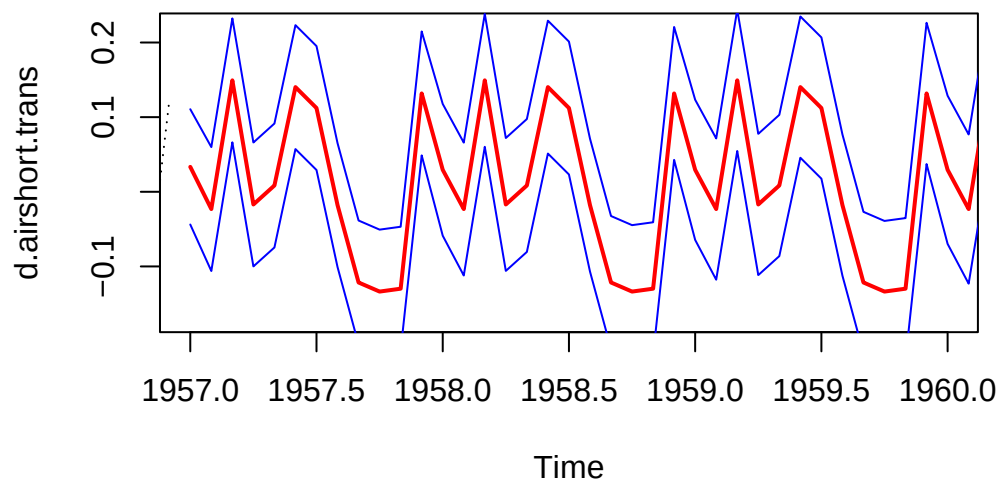
#Perform forecast
fc <- predict(fit, n.ahead=4*12)

#Plot everything
plot(
```

```

x=d.airshort.trans,
lty=3,
xlim=c(1957,1960)
)
lines(d.airshort, lwd=1)
lines(fc$pred, lwd=2, col="red")
lines(fc$pred+fc$se*1.96, col="blue")
lines(fc$pred-fc$se*1.96, col="blue")

```



Series 11

Exercise 11.1

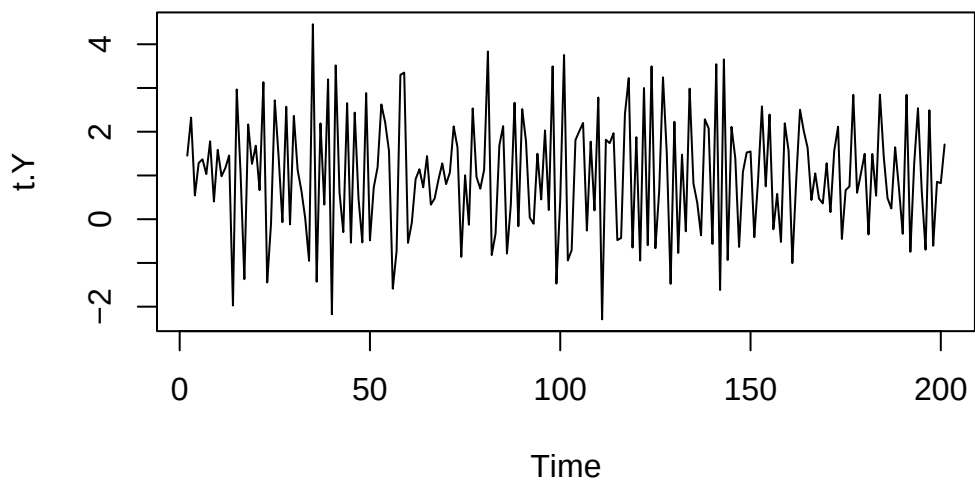
A

Process	Simplified Formula	Mean	Variance	Stationary?	Time-Series Classification
i) X_t	$t + E_t$	t (<i>Changes with time</i>)	σ^2 (<i>Constant</i>)	No	Trend-Stationary (Raw process is non-stationary)
ii) Y_t	$1 + E_t - E_{t-1}$	1 (<i>Constant</i>)	$2\sigma^2$ (<i>Constant</i>)	Yes	Stationary MA(1) process with a drift
iii) Z_t	E_t	0 (<i>Constant</i>)	σ^2 (<i>Constant</i>)	Yes	Stationary White Noise

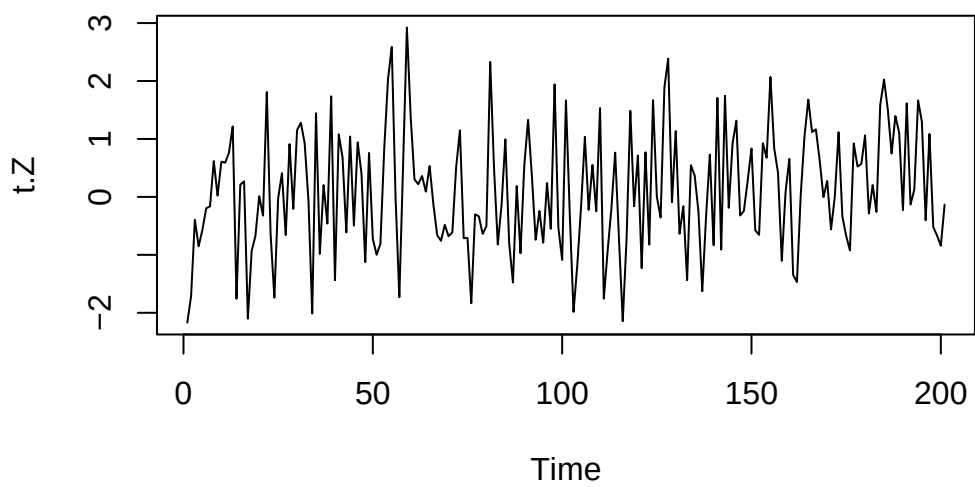
C

```
t.E <- ts(rnorm(201))
t.X <- (1:201) + t.E
t.Y <- diff(t.X)
t.Z <- t.E
```

```
plot(t.Y)
```



```
plot(t.Z)
```



Series 12

Exercise 12.1

```
library(AER)
data("Guns")
```

A

I

```
summary("Guns")
```

Length	N.unique	N.blank	Min.nchar	Max.nchar
1	1	0	4	4

II

```
# verify that panel is balanced
years <- length(levels(Guns$year))
states <- length(levels(Guns$state))
years * states == nrow(Guns)
```

```
[1] TRUE
```

The dataset is balanced!

B

I

```
library(plm)

model.se <- plm(
  formula = log(violent) ~ law,
  data = Guns,
  index = c("state", "year"),
```

```

    model = "within"
  )
model.se

```

Model Formula: $\log(\text{violent}) \sim \text{law}$

Coefficients:

lawyes
0.11366

II

```
coeftest(model.se, vcov. = vcovHC, type = "HC1")
```

t test of coefficients:

	Estimate	Std. Error	t value	Pr(> t)
lawyes	0.113663	0.035689	3.1848	0.001488 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

III

```

model = plm(log(violent) ~ law, data = Guns, index = c("state", "year"), model = "pooling")
pFtest(model.se, model)

```

F test for individual effects

data: $\log(\text{violent}) \sim \text{law}$
 $F = 260.5$, $df1 = 50$, $df2 = 1121$, $p\text{-value} < 2.2e-16$
 alternative hypothesis: significant effects